

Dendritic morphology as a dictionary for local credit assignment

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Abstract

Dendritic trees compartmentalize synaptic inputs, but which spatial distinctions a learning signal must preserve is unclear. We describe delivered credit as combinations of spatial patterns constrained by morphology, and compare tasks that place different demands on those patterns. Image classification benefits mainly from neuron-specific feedback, contextual conflict and opposing branch tuning favor routed credit, including in conductance trees, while hierarchical distractors favor ancestry at an intermediate bandwidth and input-spectrum-matched interaction tasks develop different spatial credit spectra and tolerance of fixed profiles. Reconstructed arbors provide modeled spatial capacity beyond a common broadcast, while focal shunts act as gains on a baseline-weighted ancestry partition, with spatial selectivity depending strongly on electrotonic state. Measured visual responses show no preferential ancestry alignment, delimiting the biological evidence for a framework that links task structure to useful credit resolution without assuming that finer feedback or greater dendritic depth is always advantageous.

Introduction

Learning must link a synapse’s local activity to its contribution to a circuit outcome. Cable filtering, nonlinear integration and spatially organized inhibition make locations within a dendritic tree functionally distinct [1–7]. These differences expand computation, but when does branching also help distribute useful learning signals?

Credit assignment asks how changing a synapse would affect task error. Backpropagation (BP) answers this by calculating the loss gradient—the sensitivity of task error to each parameter [8]; related sensitivity calculations apply to recurrent and steady-state networks [9, 10]. For a point neuron, each incoming-weight gradient is the product of a locally available sensitivity, called eligibility, and a neuron-specific learning signal. Hebbian plasticity captures the association between local pre- and postsynaptic variables [11], while three-factor rules combine that eligibility with an outcome-dependent signal [12]. Here, a rule is local when a synapse uses its own eligibility and an available learning signal. This definition does not require that the signal itself be computed locally.

Biologically motivated learning theories differ in how they obtain such signals. Feedback alignment communicates errors through separate pathways [13, 14]; equilibrium, predictive and prospective methods infer target states or errors from dynamics [15–17]; eligibility propagation combines temporal traces with neuron-specific signals [18]. One number shared across neurons carries less task information than separately assigned neuronal signals, although each synapse still has its own eligibility [19–21]. Recent causal evidence for distinct instructive signals across neuronal populations motivates asking what additional distinctions are useful within a neuron’s arbor [22].

We describe this organization using a *route dictionary*: a set of spatial patterns for delivering learning signals. One pattern broadcasts throughout an arbor; several can distinguish subtrees; a complete set can represent every compartment’s error. A pattern specifies the relative signal at each location, and its coefficient sets the amount delivered on a given trial. Their linear combinations give the available fields. We use *basis* for linearly independent patterns and *dictionary* when patterns may be redundant. Morphology constrains the available patterns; the task and teaching mechanism determine which combinations are useful.

Conductance changes the gains of anatomical delivery routes. Excitatory and inhibitory contacts open nonnegative conductances whose reversal potentials determine current; inhibition can lower input resistance and attenuate other

inputs even when its own current is small [23–27]. Rescaling a delivery channel preserves its spatial profile, whereas altered cable transport can reshape how local voltage affects somatic output and task error.

Dendritic learning theories already provide mechanisms for somatic prediction, segregated basal and apical teaching, interneuron-mediated errors, burst-dependent plasticity and prospective dynamics [28–34]. Detailed cable-learning models use synaptic sensitivities and local approximations to optimize a reconstructed neuron’s input–output function [35]. Our question concerns the resolution and geometry of feedback over a whole arbor: when does a morphology-constrained dictionary preserve the distinctions a task requires, and when is a simpler delivery rule sufficient?

The conductance-tree gradient factorization and LocalCA rule were introduced in our predecessor preprint [36]. Here we examine how task structure changes the spatial distinctions that credit must preserve, and connect those distinctions to anatomical routes and conductance-dependent transport.

We compare the value of preserving neuronal identity, branch selection, ancestry grouping and example-dependent credit across interacting branches. Morphology supplies candidate patterns and conductance regulates their expression. We establish forward compatibility before interpreting learning gaps; measured responses test the anatomical proposal. The question is which distinctions useful credit must preserve, rather than whether finer feedback is always better.

Results

Exact gradients separate local sensitivity from downstream credit

A useful synaptic update combines local sensitivity with the effect that the change would have on task error. For a point neuron $y_u = f_u(\sum_i w_i x_i + b_u)$, the chain rule gives $\partial \mathcal{L} / \partial w_i = x_i f'_u \delta_u$, where x_i and w_i are synaptic input and weight, b_u is bias, and $\delta_u = \partial \mathcal{L} / \partial y_u$ is the neuronal error. The derivative f'_u is evaluated at the current input sum. Thus local eligibility $x_i f'_u$ and task-derived credit supply different information (Fig. 1A).

We replace the point neuron by a rooted tree with spatially distinct addresses, with soma $n = 0$, compartment voltage V_n , and nonnegative synaptic conductances g_i . A contact with reversal potential E_i^{rev} and nonnegative activity x_i contributes current $g_i x_i (E_i^{\text{rev}} - V_n)$. In the directed artificial trees, child outputs drive their parents and a terminal-to-soma pass computes the steady state. Total input resistance R_n^{tot} is the reciprocal of the summed leak, excitatory, inhibitory and child-coupling conductances (Methods). Differentiating the current balance yields

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\delta_{n,u}^V}_{\partial \mathcal{L} / \partial V_n}. \quad (1)$$

Both excitatory (E) and inhibitory (I) contacts obey this factorization (Fig. 1A). Eligibility tells how conductance changes local voltage; compartment credit tells how that voltage changes task loss.

Here voltage error is a loss derivative, not a voltage difference. In a directed tree, each compartment has one path to the soma. Its exact voltage error is

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad (2)$$

Here $f'_i(V_i)$ is the child compartment’s output derivative, $g_{i \rightarrow k}^{\text{den}}$ is its coupling conductance to parent k , and $\delta_{0,u}^V = f'_0(V_0) \delta_u$ is the somatic voltage error. Thus the path multiplier $\tilde{\alpha}_n$ converts one somatic error into a spatial field. The conductance-only component of this product is attenuated by additional conductance on a path’s ancestors; branch nonlinearities and forward voltages can change as well (Supplementary Section S1).

Reciprocal cables instead require a joint adjoint calculation of how local currents affect task loss (Methods). These mathematical sensitivities do not establish a biological error-transmission mechanism.

Our local credit assignment (LocalCA) rule retains the local sensitivity but supplies an approximate learning signal to each compartment. A hat denotes this delivered field:

$$\widehat{\nabla}_{g_i} \mathcal{L} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \widehat{\delta}_{n,u}^V, \quad (3)$$

$$\widehat{\delta}_u^V = A_u \mathbf{c}_u. \quad (4)$$

Columns of $A_u \in \mathbb{R}^{N_u \times K}$ are fixed spatial delivery profiles over N_u routed nonsomatic compartments; $\mathbf{c}_u$ contains their example-dependent coefficients. Broadcast, subtree and compartment profiles give increasingly detailed spatial addresses (Fig. 1B). This linear delivery model does not follow automatically from the forward cable equations and does not specify how a circuit estimates the coefficients.

At $K = 1$, the all-ones column $A_u = \mathbf{1}$ broadcasts one coefficient through the arbor; at $K = N_u$, the identity matrix $A_u = I$ can reproduce the full nonsomatic field. Nested subtrees provide intermediate dictionaries. Profile count, rank (the number of independent profiles) and coefficient information are distinct resources. Exact path transport combines one somatic scalar with state-dependent local derivatives; it does not require an independent external error for every compartment.

Capture is the fraction of squared learning-signal magnitude that a dictionary can reproduce. For a nonzero field δ ,

$$\mathcal{C}(A; \delta) = \frac{\|P_A \delta\|^2}{\|\delta\|^2}, \quad P_A = AA^\dagger.$$

Here P_A gives the best least-squares reconstruction from the profiles; $A^\dagger$ is the Moore–Penrose pseudoinverse and $\|\cdot\|$ the Euclidean norm. Thus capture is a fraction of squared field magnitude, not a fraction of correct decisions. The mean of these per-field ratios weights fields equally. In contrast, $\text{tr}(P_A C) / \text{tr}(C)$, with $C = \mathbb{E}[\delta \delta^T]$, is the fraction of total ensemble energy captured, weighting fields by their energy. Here C is the uncentered second-moment matrix and tr denotes its trace, the sum of diagonal entries. The two capture measures need not agree. Projection chooses the best coefficients in the specified metric; an implemented learning rule need not supply them.

The expansion applies in a specified coordinate. In the nonlinear image models, we route errors of compartment outputs and retain the local output derivative in eligibility; voltage-error capture is reported separately. Identity-output conductance trees use voltage errors directly. Supplementary Table S14 compares the coordinates, coefficient sources and state dependence. Dictionary capacity answers what fields can be delivered. Learning experiments determine which of those distinctions improve task performance.

One signal per neuron captures most of the image-task learning benefit

We first tested credit resolution on image classification. All image benchmark networks used one dendritic layer and a trainable linear readout. In the primary comparisons, the readout supplied exact neuronal output errors before the feedback intervention. The MNIST comparison used 128 directed [3, 3] trees, each with three proximal and nine terminal compartments, receiving flattened images. Shunting and raw-additive architectures shared the graph, initial contact construction, nonlinearities and training protocol, with architecture-specific initialization. The additive control removed reversal-potential driving forces and the conductance denominator, retaining signed current summation without shunting.

Six rules separated neuronal identity, spatial resolution and learning in the dendritic core (Fig. 1B,C). A strict scalar shared one signal across neurons; neuron-specific feedback broadcast each neuron’s own output error through its arbor. A projected one-profile rule averaged the exact activation errors over all twelve nonsomatic compartments, while three subtree profiles averaged them within each proximal compartment and its three children. Both projections acquired coefficients from the same exact field and preserved its common-mode mean at identical parameters, although their field norms could differ. Exact transport retained every compartment error. A decoder-only reference trained the readout while freezing the core. Three development seeds selected each rule’s rate from three common multipliers; ten fresh paired seeds per architecture evaluated selected and original common rates over 180 epochs (Methods; Supplementary Fig. S49A–D).

At selected rates, neuron-specific feedback exceeded the strict scalar by 9.23 percentage points in shunting trees and 7.46 points in additive trees, reaching 97.12% and 97.11% accuracy. Decoder-only learning reached 80.96% and 89.57%, respectively (Fig. 1C). Increasing the projected dictionary from one to three profiles changed accuracy by only 0.049 points in shunting trees (95% paired interval, -0.048 to 0.141) and 0.036 points in additive trees (-0.056 to 0.122 ; Fig. 1D). The common-rate comparison gave similarly small differences (Supplementary Fig. S49E). Exact transport added 0.010 points over neuron-specific feedback in shunting trees, but 0.182 points in additive trees (0.114 – 0.252 ; positive in all ten seeds). Both exact transport and one projected profile reached about 97.3% in additive trees. Most learning benefit therefore came from preserving neuronal coordinates; the smaller additive improvement was attainable without separating subtrees.

The original fifteen-seed cohort and exact-gradient checks remain in Supplementary Fig. S45. Its trained fields retained spatial structure: moving from one to three profiles raised mean voltage-credit capture from 0.480 to 0.625 in shunting trees and from 0.519 to 0.718 in additive trees. The new checkpoint analysis likewise found improved capture without a corresponding accuracy gain from subtree distinctions (Supplementary Fig. S49F). It reports activation-space capture, matching the trained projections, with voltage-space results supplied separately. These reconstructions measure available capacity, not the quality of a biological coefficient encoder. Separate recipient-permutation controls also reduced accuracy while preserving signal values and bandwidth, implicating neuronal identity (Supplementary Fig. S45F,G).

The pattern extended beyond the primary readout. With fixed random soma-level feedback, neuron-specific signals improved over scalar feedback by about eleven points, while exact paths added at most 0.09 points (fifteen paired seeds; Fig. 1E; Supplementary Fig. S30). Fashion-MNIST also favored neuronal feedback. On flattened CIFAR-10, exact paths reduced accuracy relative to neuronal feedback by 0.86 points (0.55 – 1.17), while remaining equivalent to matched backpropagation within the predefined margin (Fig. 1E; Supplementary Fig. S4).

These benchmarks show that a spatially structured error field need not confer a learning advantage on finer delivery. Supplementary Figs. S4 and S8 retain further depth and task-generator controls. A one-step smoothness bound explains how restricting noisy updates can improve expected progress, but its initialization utility lost to a fixed maximum-budget tree in all twenty seed averages of a 12,800-fit prospective test, in both selection arms. The bound therefore concerns local update quality rather than optimal morphology selection (Methods; Supplementary Sections S3–S4). We next introduce task structure that makes particular spatial distinctions useful.

Branch-selective information prevents updates from inactive distractors

To isolate branch-selective information, we used one logistic output with $B \in \{2, 4, 8\}$ parallel branch weight vectors and no internal forward compartments or conductance nonlinearity (Fig. 2A). Context selected the branch determining the output, while every branch received an image and formed an ungated local eligibility.

Inputs were average-pooled, standardized Fashion-MNIST images from classes 0 and 6. The selected image determined the binary target. At conflict dose $\chi = 0$, all branches received that image; increasing χ independently replaced nonselected inputs with opposite-class images. For example i , let b_i^* be the selected branch, $\mathbf{x}_{i,b}$ its input vector and $y_i \in \{0, 1\}$ the target. The branch weights $\mathbf{w}_b$ give the pre-probability score (logit) and output error

$$z_i = \mathbf{w}_{b_i^*}^T \mathbf{x}_{i,b_i^*}, \quad \delta_i = \sigma(z_i) - y_i. \quad (5)$$

Here $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function. The exact derivative of every unselected branch is zero, irrespective of its input. Increasing conflict therefore does not create opposing nonzero exact branch gradients. It makes a rule that ignores the selector increasingly sensitive to irrelevant, opposite-class eligibilities.

For N examples, correctly selected and neuron-shared gradient estimates were

$$\mathbf{d}_b^{\text{selected}} = N^{-1} \sum_i \delta_i \mathbf{1}[b_i^* = b] \mathbf{x}_{i,b}, \quad \mathbf{d}_b^{\text{shared}} = N^{-1} \sum_i \delta_i B^{-1} \mathbf{x}_{i,b}. \quad (6)$$

The indicator $\mathbf{1}[\cdot]$ is one when its condition holds, zero otherwise; $1/B$ matches scale without supplying branch information. A feedback selector and a branch-local eligibility gate give the same update (Fig. 2B), testing access to selection rather than a need for independently computed external errors.

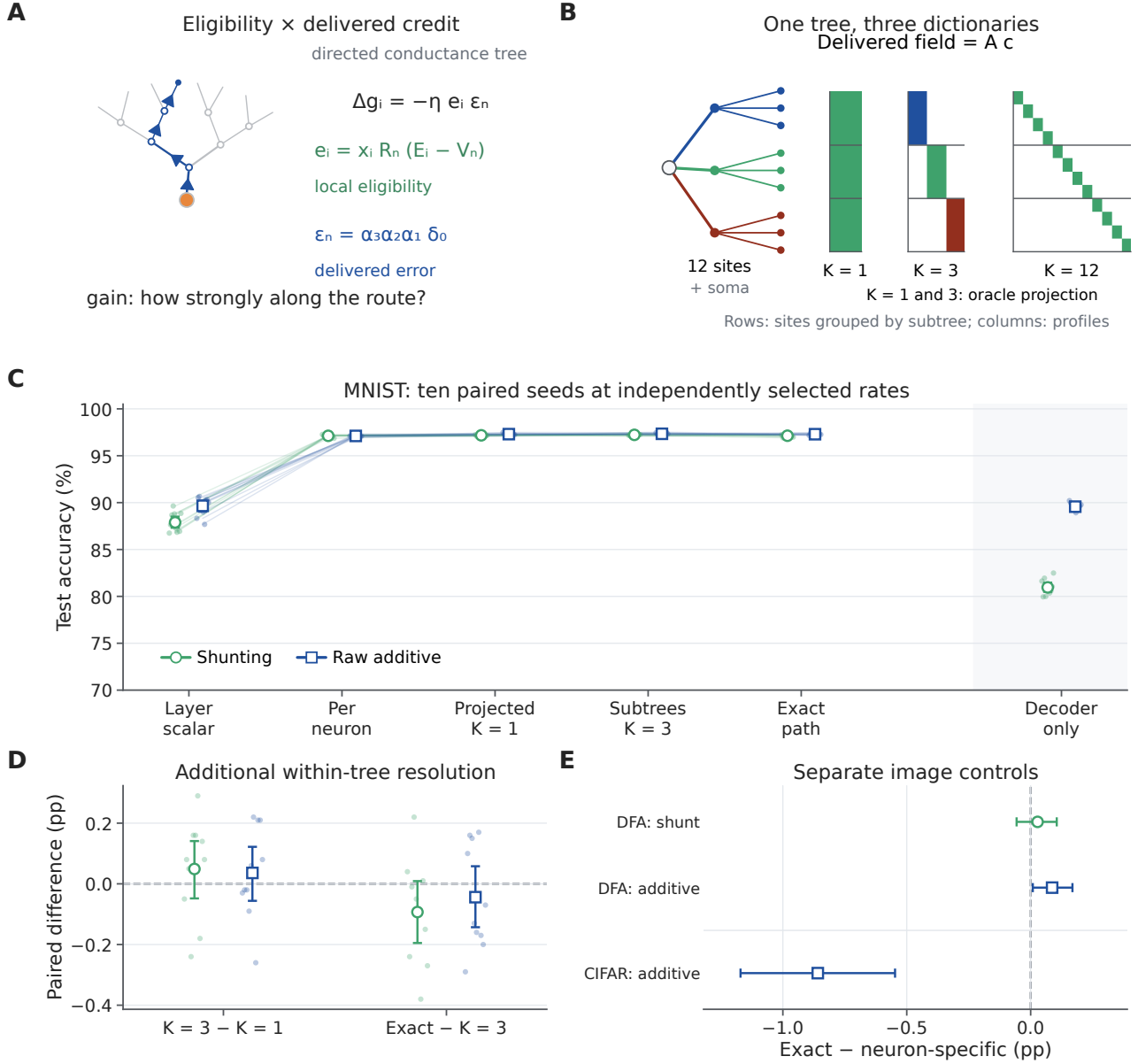


Figure 1: Task-derived credit separates neuronal identity from resolution within a dendritic tree. **A**, Conductance updates combine local eligibility e_i and delivered compartment credit ϵ_n . Here η is learning rate, x_i presynaptic activity, R_n input resistance, and $E_i - V_n$ driving force. Exact directed-tree transport multiplies somatic error δ_0 by path derivatives α ; reciprocal cables require an adjoint solution. **B**, Over the twelve nonsomatic sites, $K = 1$ broadcasts, $K = 3$ groups each proximal site with its three children, and $K = 12$ resolves every site. Projected rules retain exact somatic errors. The $K = 1$ and $K = 3$ training conditions use oracle projection of the exact activation-error field. K counts spatial profiles, not independently computed external errors. **C**, MNIST with one 128-neuron dendritic layer and a linear readout. Neuron-specific credit broadcasts each neuron's exact somatic activation error; the strict scalar removes neuronal identity. The decoder-only reference freezes dendritic parameters. Thin lines pair ten fresh seeds per architecture; large symbols show means and 95% descriptive intervals. All fits use 180 epochs, rates selected on three separate development seeds, and validation-selected checkpoints. **D**, Paired accuracy differences compare $K = 3$ with projected $K = 1$, and exact paths with $K = 3$. The projected rules use the same exact-field coefficient source and preserve the same mean at identical states, but their norms can differ. Points show seeds with paired 95% intervals. **E**, Separate legacy cohorts compare exact paths with neuron-specific broadcast under fixed random between-neuron feedback on MNIST (direct feedback alignment, DFA; 15 seeds) or a readout derivative on flattened CIFAR-10 (20 seeds). Colors and shapes follow **C**. Percentage points are abbreviated pp; original 15-seed readout-derived MNIST ladders remain in Supplementary Fig. S45. Supplementary Fig. S49 adds common-rate and capture controls.

157 A simplified calculation asks when the selected branch’s evidence is outweighed by conflicting evidence
158 sent to other branches. With centered, symmetric class evidence, the shared-mode prediction crosses zero at
159 $\chi_c = B/[2(B - 1)]$: 1, 2/3 and 4/7 for two, four and eight branches (Fig. 2C; Methods). More branches therefore
160 predict an earlier loss of compatibility. This average calculation is distinct from the conflict dose at which a trained
161 model reaches chance accuracy.

162 At zero conflict, selected and shared rules achieved similar accuracy. Their difference increased with conflict,
163 reaching 35–58 percentage points at full conflict across the tested branch counts (20 paired seeds; Fig. 2D). In every
164 seed, the advantage of correct routing increased with conflict. A cyclic reassignment sent every signal to the wrong
165 branch at matched rank and sparsity. It was harmful even at zero conflict, reducing accuracy by 12–25 points relative
166 to correct selection. Analytic backpropagation and an explicitly gated grouped-point model matched correct selection
167 exactly, as required by their algebraic equivalence (Fig. 2E; Supplementary Fig. S29).

168 The shared rule reached chance at lower conflict as branch count increased, consistent with the predicted shift
169 (Fig. 2F); five two-branch seeds never crossed in the sampled range. Theoretical, mean-curve and individual-seed
170 crossings are distinct. Gradient alignment also changed during learning (Methods; Supplementary Fig. S29D–F).
171 Credit reaching inactive branches can therefore harm learning; dendrites offer one possible selection site. A separate
172 two-stream task tests learning and interference effects (Supplementary Fig. S19).

173 **Ancestry provides a modest advantage at an intermediate feedback budget**

174 Branch selection compares fully shared with fully resolved feedback. We next asked whether nested subtree routes
175 offer a useful intermediate dictionary. The task contained eight independent, eight-dimensional terminal weight
176 blocks at the leaves of a balanced binary feedback tree. Only the context-selected block determined the logit. Internal
177 tree nodes defined feedback supports and did not perform serial forward computation (Fig. 3A).

178 The selected stream carried the class signal. Distractors carried opposing signals that strengthened with distance
179 through the feedback hierarchy, deliberately favoring task-aligned grouping. At each budget $K \in \{1, 2, 4, 8\}$, the
180 routing field had K independent patterns and unit-length rows. Ancestry routes grouped the terminals into the
181 whole tree, halves, sibling pairs or individual terminals. Permuting task streams across leaves changed alignment
182 without changing the streams or forward parameter budget. Here bandwidth counts independent feedback patterns,
183 not physical stages or independent scalar losses; other single-pattern controls need not be uniform. The complete
184 generator and update are given in Methods.

185 The generator predicts where an intermediate ancestry benefit can occur. The selected stream has class coefficient
186 one; its sibling, same-half nonsiblings and other-half streams have coefficients -0.15 , -0.45 and -0.75 . The
187 unnormalized sum of the class signal within the selected ancestry group gives -3.05 , -0.05 , 0.85 and 1 at
188 $K = 1, 2, 4, 8$, respectively. Thus coarse pooling reverses or nearly cancels the signal, sibling-pair routing first
189 preserves a positive intermediate-group signal, and full resolution removes the grouping distinction. The implemented
190 unit-row normalization divides these sums by the square root of group size, preserving their signs. This is an analytic
191 consequence of the fixed task design; its location is not an independently discovered optimum (Fig. 3B).

192 Controls preserved rank while changing recipient assignment, support geometry or sparsity: cyclic derangement,
193 depth-interleaving, random-sparse and dense random fields. The per-seed best-control summary takes their maximum
194 after training; it is not a learned selector. A training-informed principal-component reference is retained separately
195 (Methods).

196 Increasing bandwidth improved correct routing, while derangement caused large losses (Fig. 3C). Degree- and
197 depth-preserving rewiring removed the task-aligned advantage at intermediate budgets but did not affect fully shared
198 or fully resolved ancestry routes (Fig. 3D).

199 The ancestry-specific effect was much smaller. Ancestry performed worse than the best matched control at
200 low budgets, exceeded it by 1.27 percentage points at $K = 4$, and tied at full resolution. The paired $K = 4$ interval
201 was 0.59–1.95 points, with 15/20 positive differences; a paired Wilcoxon signed-rank test gave Holm-adjusted
202 $P = 0.0101$ across four budgets. Individual control contrasts and paired seed differences distinguish this effect from
203 the much larger correct-versus-deranged contrast (Fig. 3E,F).

204 Flat, grouped-point and gated-point representations receiving the same field behaved identically by construction.

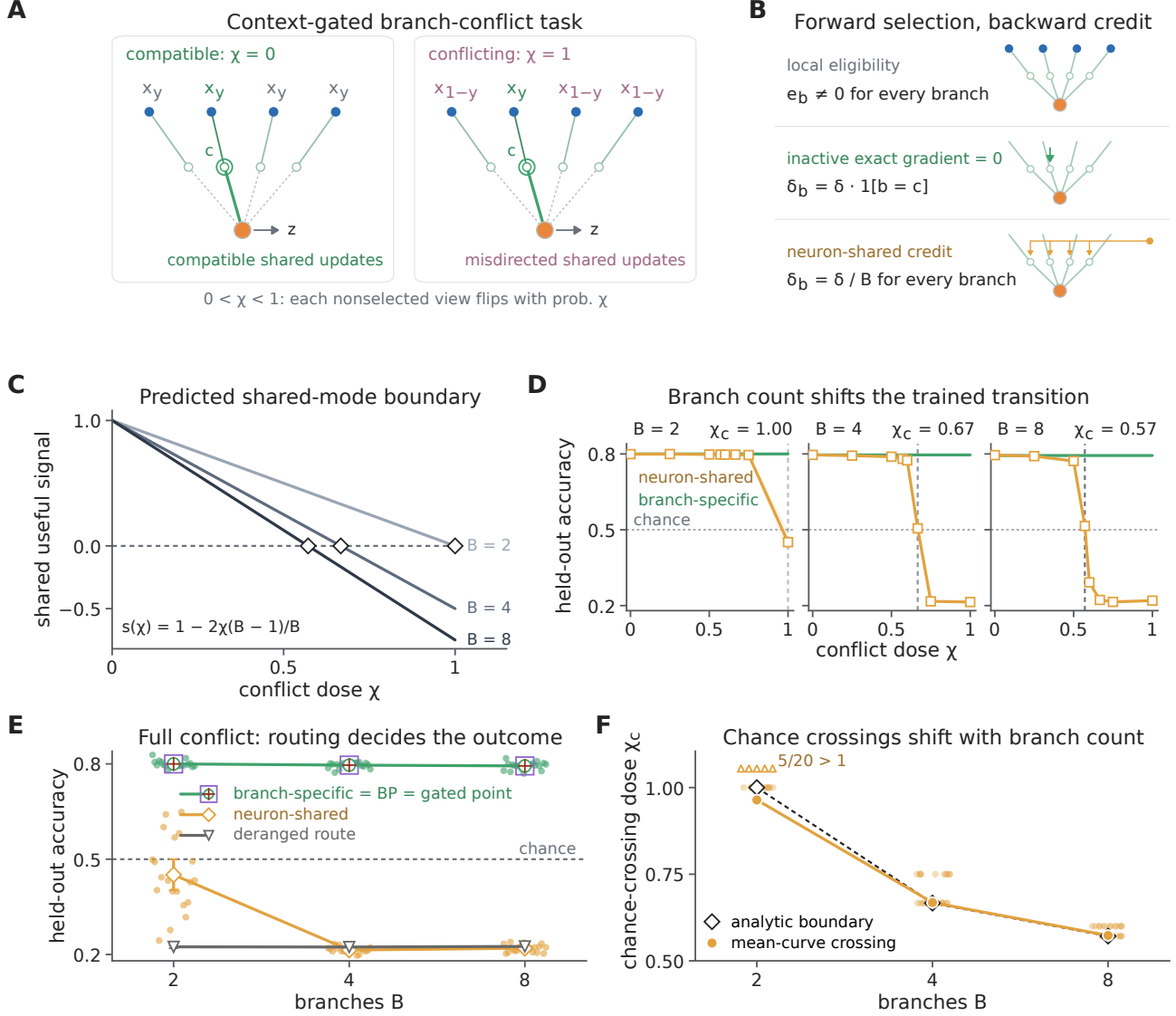


Figure 2: Branch selection prevents learning from inactive distractors. One logistic output combines B parallel branch weight vectors; there are no serial or conductance nonlinearities. **A**, Context-gated Fashion-MNIST task. Context c selects the branch determining logit z ; every branch receives an image and forms an ungated eligibility. Conflict dose χ independently replaces each nonselected image by an opposite-class image. Solid and dashed proximal paths are selected and gated-off forward routes; the double ring marks context. **B**, Selected feedback: $\delta_i \mathbf{1}[b = b_i^*]$, shared feedback: δ_i / B . Here δ_i is the output error, b_i^* the selected branch and $\mathbf{1}$ the indicator; unselected exact derivatives are zero. A local eligibility gate gives the same update. **C**, Centered symmetric-evidence prediction $s(\chi) = 1 - 2\chi(B - 1)/B$: shared-mode compatibility vanishes at $\chi_c = B/[2(B - 1)]$. This mean-field boundary is distinct from a trained chance crossing. **D**, Held-out accuracy over conflict dose. **E**, At full conflict, correct selection, backpropagation (BP) and the algebraically equivalent gated-point calculation coincide; cyclic derangement sends signals to the wrong branch at matched rank and sparsity. Selection-by-conflict interactions are positive in all 20 seeds for each B . **F**, Analytic threshold (diamonds), first interpolated crossing of mean accuracy (orange), and each seed's first sampled at-or-below-chance dose (faint dots). Open overflow triangles retain five $B = 2$ seeds that do not cross within the sweep. Curves and bars summarize 20 paired seeds with 95% seed-bootstrap intervals. Branch-count comparisons do not hold capacity fixed.

205 The experiment therefore supports a conditional advantage of ancestry-shaped feedback under limited bandwidth,
 206 without attributing the effect to dendritic material or treating equivalent representations as independent confirmations.
 207 Learning the route coefficients was another limitation. In twenty fresh seeds, an estimator trained on separate
 208 noisy context cues improved accuracy over a frozen profile by 7.16 points, but remained 54.47 points below oracle
 209 context coefficients. Exploratory selection of its most probable route narrowed the gap, whereas delaying cues
 210 removed the benefit (Supplementary Fig. S32; Methods). These tests expose requirements for selective, timely
 211 delivery; they do not derive the context signal from local eligibility.

212 Cross-branch interactions change the spatial credit that learning uses

213 Interactions across branches can change credit’s sign and magnitude between examples. We compared targets with
 214 identical input-sensitivity spectra on the same representationally compatible tree, separating credit delivery from
 215 forward capacity.

216 Eight independent inputs x_i were equally likely to be -1 or $+1$. Each branch computed $u = a + bu_L + cu_R + du_Lu_R$
 217 with four trainable coefficients. This multi-affine unit is affine in either child while the other is fixed. Pairwise
 218 targets summed four disjoint products, $f_M = \frac{1}{2} \sum_{(i,j) \in M} s_{ij} x_i x_j$; quartic targets combined complementary quartets,
 219 $f_A = \frac{1}{2} (s_A \prod_{i \in A} x_i + s_{\bar{A}} \prod_{i \notin A} x_i)$. Here M partitions inputs into pairs, A is one quartet, $\bar{A}$ its complement, and signs s
 220 were independently sampled. Within each seed, targets shared a compatible balanced tree, input assignment, initial
 221 weights and examples (Fig. 4A). Both have input-sensitivity second moment $\mathbb{E}[\nabla_x f \nabla_x f^T] = I_8/4$, giving identical
 222 strengths of independent sensitivity directions.

223 Compatibility has an exact criterion. At each subtree boundary, arrange interaction coefficients into a matrix
 224 indexed by products of inputs inside and outside the subtree. Removing the constant inside row must leave rank at
 225 most one, because only one scalar leaves the branch. This condition is necessary and sufficient across all boundaries;
 226 discarded singular values bound representation error (Methods; Supplementary Section S4). It constrains forward
 227 capacity, not the learning rule. Boolean examples and the structure census appear in Supplementary Figs. S36–S37
 228 and S43–S44.

229 All credit rules used the same exact local eligibilities and scalar output error. We compared exact paths with
 230 unit broadcast and a fixed six-site profile calibrated from each student’s mean path derivatives over 256 unlabeled
 231 initialization examples. Three development seeds selected learning rates with equal budgets; twenty fresh paired
 232 seeds tested the frozen choices. A separate nested target summed products over prefixes of two, four, six and eight
 233 inputs on its own compatible tree.

234 After 1,024 Adam steps, exact and calibrated credit gave held-out normalized mean squared errors (NMSE) of
 235 0.0317 and 0.0236 for pairwise targets, 0.1516 and 0.9813 for quartic targets, and 0.0230 and 0.7690 for nested targets
 236 (Fig. 4B–D). NMSE divides squared prediction error by clean target variance. The predefined quartic-minus-pairwise
 237 difference in calibrated-minus-exact gaps was 0.838 (95% paired seed-bootstrap interval, 0.719–0.931), positive in
 238 all twenty seeds; a common rate gave 0.843 (0.727–0.935; Fig. 4E). Some exact quartic fits retained appreciable
 239 error at this fixed budget. Equal label noise gave different NMSE floors because pairwise and quartic target variances
 240 were one and one-half.

241 The trained fields reveal a distinction missed by the input spectrum. Let $\mathbf{q}(x)$ contain the six nonsomatic
 242 derivatives of root output with respect to branch output; exact branch credit is scalar output error times $\mathbf{q}$. Allowing
 243 an oracle amplitude per example, the best fixed profile captured 99.7% of exact-trained path-field energy for pairwise
 244 targets, but 46.1% for quartic and 44.8% for nested targets (Fig. 4F). Loss-error weighting, eligibility weighting,
 245 site-normalized spectra and replay of 600 earlier fits preserved this separation (Supplementary Section S4). Uniform-
 246 profile capture did not: a nearly one-dimensional field can have a signed, nonuniform direction. Positive fixed-profile
 247 scaling largely cancels in Adam, explaining calibrated/sign-only agreement; it does not explain the slower pairwise
 248 learning with changing exact paths.

249 Thus input-matched targets developed different credit spectra and tolerance of fixed profiles. At fixed weights
 250 and inputs, however, path derivatives are target-independent: their spectrum describes learned states and coordinates,
 251 not a task-only initialization diagnostic. Failure of the tested profiles supports example-dependent spatial credit
 252 without establishing that six independent channels or exact gradients are necessary.

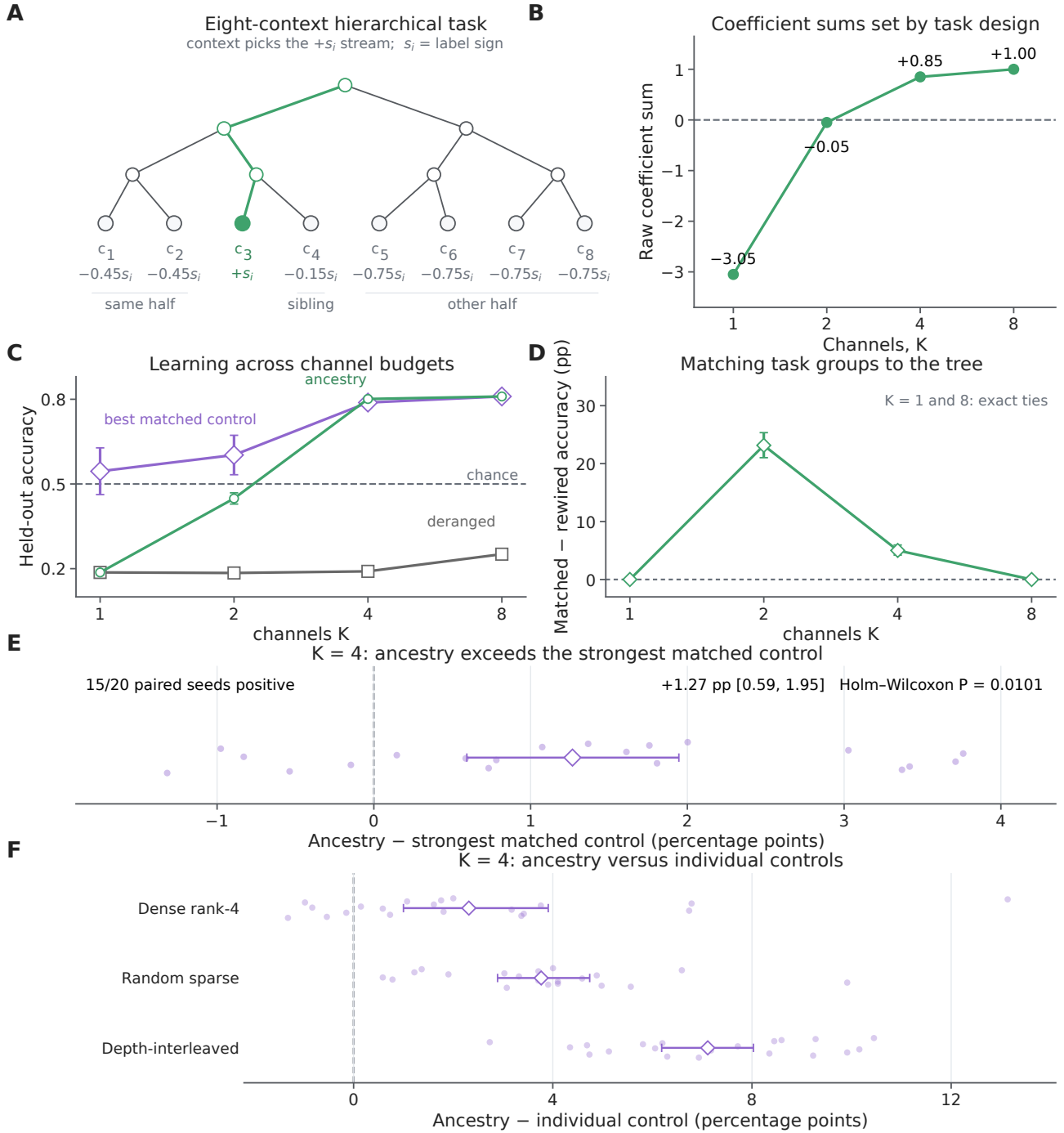


Figure 3: Hierarchical distractors favor ancestry at intermediate bandwidth. **A**, Eight streams carry label sign s_i along noisy teacher features. The selected stream has coefficient one; sibling, same-half and opposite-half distractors have coefficients -0.15 , -0.45 , -0.75 . **B**, Coefficient sums in the cued ancestry group at $K = 1, 2, 4, 8$. Implemented feedback divides these sums by square root of group size, preserving their signs. The four-route sign change follows from the design; it is not an independently discovered accuracy optimum. **C**, Held-out accuracy for correct ancestry, derangement and the per-seed maximum over four matched controls. This maximum is calculated after training, not learned by a selector. Dashed line: chance. **D**, Task-matched-minus-rewired accuracy at equal degree and depth. **E**, The four-route ancestry-minus-best-control effect, 1.27 pp (95% interval, 0.59–1.95), with 15/20 positive seeds and Holm-adjusted two-sided Wilcoxon $P = 0.0101$ across four budgets. **F**, Individual dense, sparse and depth-interleaved contrasts on a separate scale. The large derangement effect is shown in **C**. Symbols and bars are means and 95% seed-bootstrap intervals; small points in **E,F** are twenty paired seeds. Endpoint ties follow from coincident operators.

Additional controls delimit the result. Fixed-profile Adam learned a compatible positive-conductance teacher accurately; broadcast learned aligned XOR-of-AND, whereas parity favored exact credit under the tested optimizer (Supplementary Figs. S42–S44). Noisy-query interaction estimation also yielded useful trees, retained as supporting structure-design evidence in Supplementary Figs. S41–S42. Interaction structure therefore motivates specific credit comparisons rather than a universal monotonic resolution law.

Context-dependent branch tuning makes spatial credit useful in conductance trees

We next tested how branch tuning affects credit demand in a conductance tree. A first seven-compartment construction with monotonic branch responses was learned accurately by all tested rules. Under Adam, mean broadcast NMSE remained below 0.0006 after extended training despite a varying path field (Supplementary Fig. S50). We therefore tested a separate construction in which context selected branches with either aligned or opposing feature tuning (Fig. 5A,B).

Four terminal compartments fed two proximal compartments and a soma, using the directed E/I conductance balance (Methods). Both subtrees received the same positive feature pairs, $\exp(z_j)$ and $\exp(-z_j)$, for two independent latent inputs z_j . A binary context increased inhibitory conductance in one proximal compartment. Teacher excitation and inhibition produced increasing feature responses in both subtrees for the aligned task, but increasing responses in one and decreasing responses in the other for the opposed task. Teacher and student shared the circuit, ensuring representability. All 24 student conductances were plastic; paired tasks and credit rules shared generic, independently perturbed initial weights and example streams.

We compared exact credit, uniform broadcast, a fixed profile calibrated at initialization, and three fixed spatial patterns. Two patterns addressed the terminal descendants of the respective proximal branches; the third addressed the proximal pair. Their example-dependent coefficients were least-squares projections of the exact field, making this an oracle test of the dictionary’s delivery capacity. Local synaptic and coupling eligibilities were unchanged across rules.

With equal learning-rate tuning, 20 fresh paired seeds showed a large task-dependent gap (Fig. 5C,D). At validation-selected states within 4,096 Adam updates, mean held-out NMSE on the opposed task was 0.000020 with exact credit and 0.664 with the calibrated profile, whereas both learned the aligned task accurately. The opposed-minus-aligned difference in the calibrated-minus-exact gap was 0.664 (95% paired interval, 0.511–0.804). Continuing every condition to 16,384 updates retained a contrast of 0.628 (0.464–0.774), positive in all 20 seeds (Fig. 5E). The three-pattern dictionary remained close to exact credit (Fig. 5C,D). SGD and restarts with much wider conductance bounds gave the same qualitative result (Supplementary Fig. S51). A six-rate, budget-matched development sweep retained large broadcast errors (Supplementary Fig. S52).

The gradient estimates reveal competing demands. At the same broadcast-trained weights, conditional mean gradients from the two contexts cancelled more strongly under broadcast than under exact credit (Fig. 5F). Individual-example credit gradients still had nonnegative alignment with the exact gradient. Context-dependent branch selection therefore improved learning in a conductance circuit by preserving distinctions lost when gradient estimates were averaged. This result concerns a binary-context task and three supplied spatial patterns; it does not establish a general relationship between path-field rank and required feedback resolution.

Serial computation helps learning, while credit benefits depend on budget and metric

The interaction comparison holds a compatible forward tree fixed while changing credit. We next varied serial forward structure itself, asking whether successive compartments help remove unwanted multiplicative changes in input strength. Physical depth D_p counts forward stages, whereas feedback resolution counts distinctions in the backward field.

Each network contained 64 somata and eight nonsomatic compartments per soma. D1 [8], D2 [2, 3] and D3 [2, 1, 2] distributed those compartments across one to three stages while preserving contacts and trainable resources. Branching vectors are ordered from soma to tips. A grouped-point control connected all eight modules directly to the soma; raw-additive trees removed divisive conductance normalization. Flexible point networks provided additional,

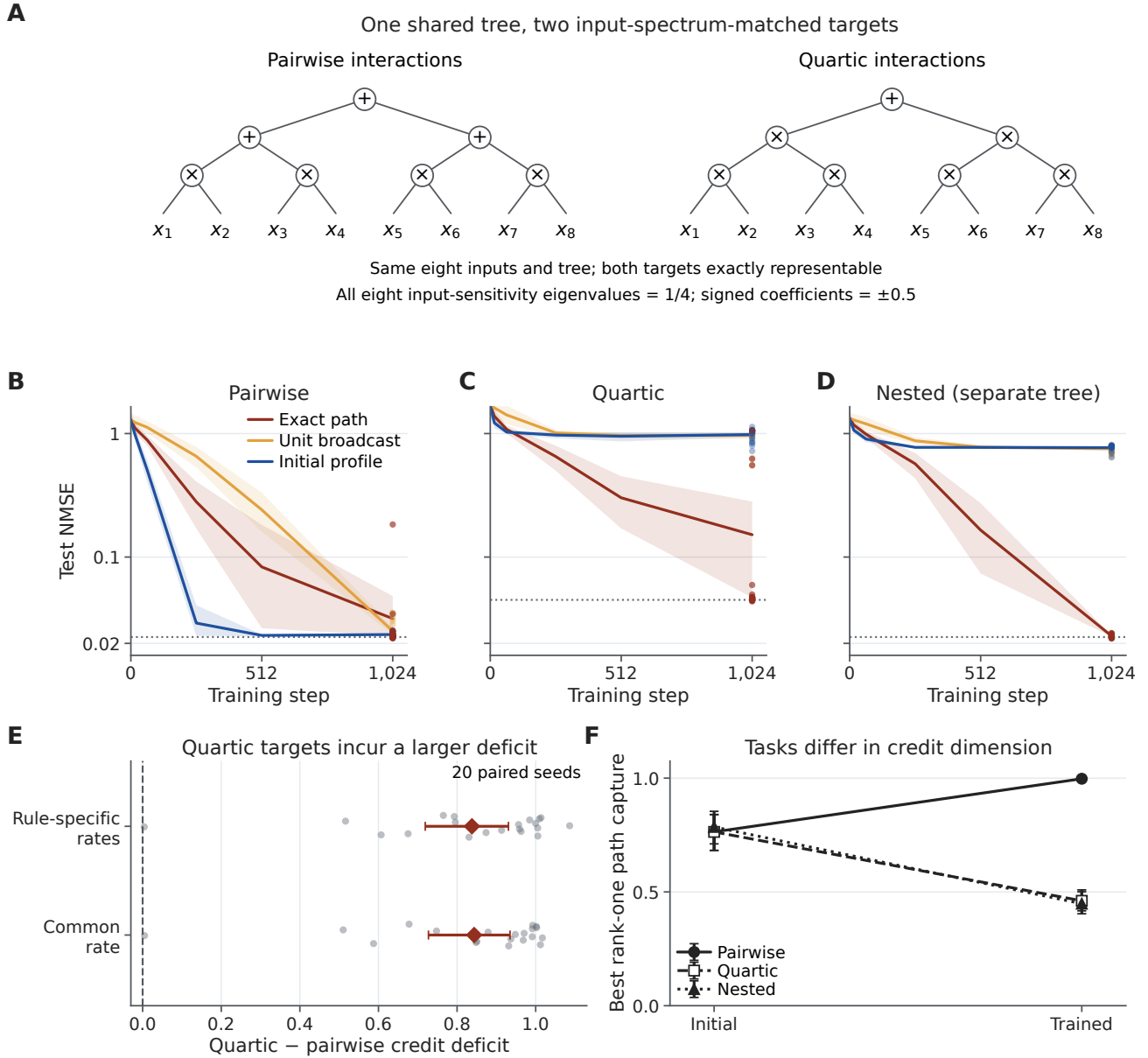


Figure 4: **Input-spectrum-matched tasks differ in learned credit geometry and tolerance of fixed spatial credit.** **A**, Pairwise and quartic targets share eight inputs, a balanced tree, initialization and example streams. Drawings show interaction supports; coefficients have random signs and magnitude 0.5, with input labels permuted together within each seed. Both targets are exactly representable and have input-sensitivity second moment $I_8/4$. **B–D**, Held-out NMSE during Adam training with exact path credit, unit broadcast and an initially calibrated profile. The latter averages the initial path derivatives on 256 unlabeled examples and remains fixed. The somatic derivative is one for all rules. Nested targets use a separate compatible tree. Rates were selected using three development seeds per rule. Lines and bands show means and descriptive 95% whole-seed bootstrap intervals; endpoint dots retain all twenty fresh seeds. Dotted lines indicate noise-only NMSE: 0.0225 for pairwise and nested, 0.045 for quartic targets. **E**, Quartic-minus-pairwise difference in the calibrated-minus-exact NMSE gap, at rule-specific rates and a common rate of 0.003. **F**, Best rank-one capture of six-site path-field energy before and after exact-credit training. The spatial direction and example-specific amplitudes are oracle fits, not an implemented encoder. Points and intervals in **E,F** summarize twenty paired seeds. All fits stop at 1,024 steps.

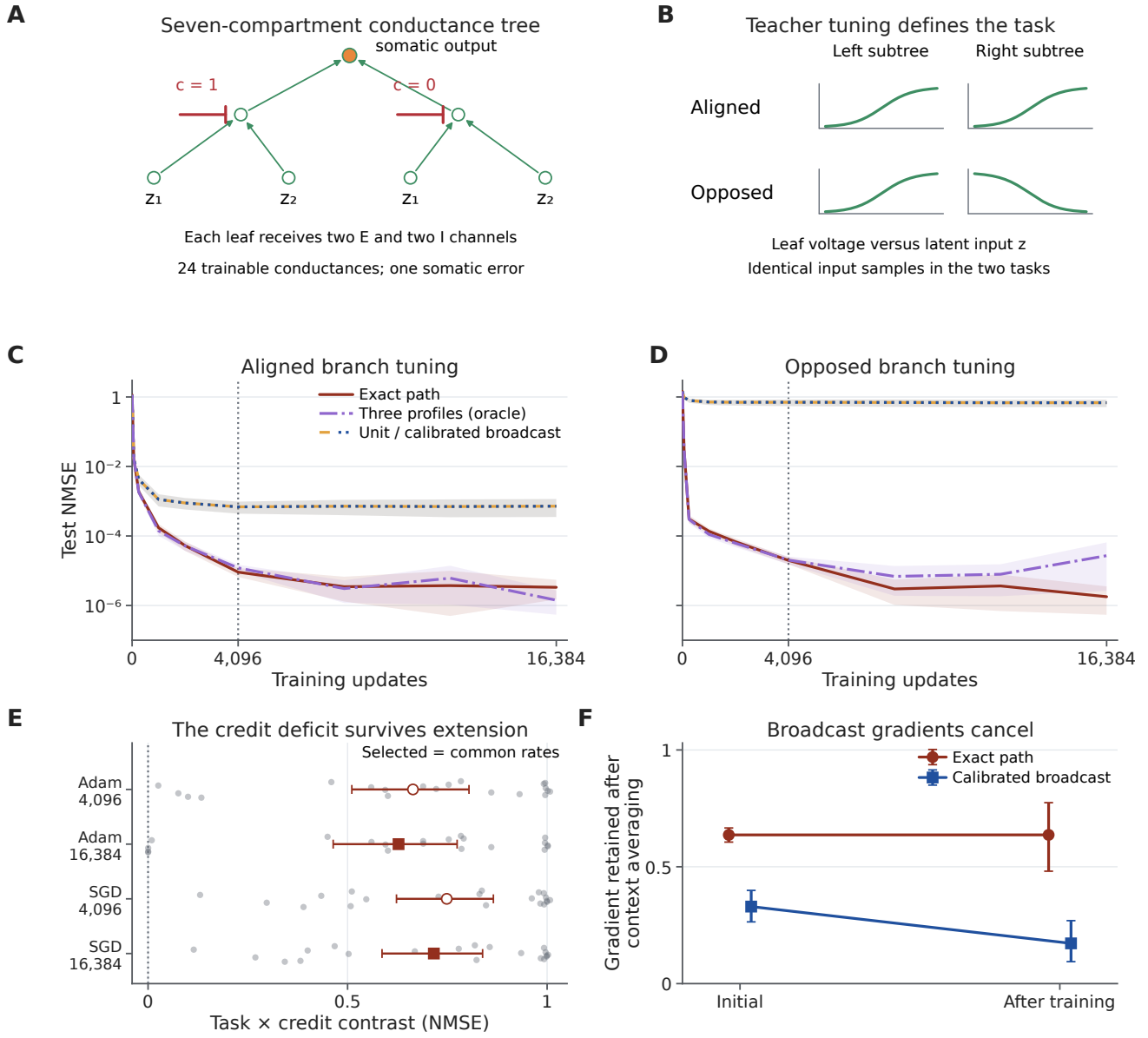


Figure 5: Task-dependent credit resolution in a conductance tree. **A**, Directed tree with four terminal compartments, two proximal compartments and one soma. All 24 conductances are trainable. Two excitatory and two inhibitory channels at each leaf receive activities $\exp(\pm z_j)$; the same latent features z_1, z_2 reach both subtrees. Binary context c sets proximal inhibition. **B**, Nominal leaf voltage versus latent input. Teacher E/I tuning produces aligned or opposed branch responses with identical input samples. **C, D**, Adam test NMSE at fixed checkpoints in 20 paired seeds. The dotted line marks 4,096 updates; all fits continue with unchanged optimizer state to 16,384. Near-overlapping unit and calibrated broadcast curves share a legend entry. The three-profile control uses oracle projection coefficients. **E**, Calibrated-minus-exact NMSE deficit on opposed minus aligned targets, using validation-selected checkpoints within each budget. Small points show paired seeds; large symbols show means. Frozen rule-specific and common rates coincide. **F**, Fraction of distal gradient magnitude retained when the two context-specific averages are mixed. Both evaluated rules use the same initial or broadcast-trained weights, inputs and eligibilities. Trained weights are selected within the 16,384-update window; zero indicates complete cancellation and one none. Bands and whiskers in **C–F** are descriptive 95% whole-seed bootstrap intervals from 20,000 draws.

approximately parameter-matched comparisons.

Distal excitation carried class information multiplied by nuisance gains. Inhibitory sensor blocks reported those gains with alignment α : zero retained the sensor distribution but removed trial-specific information, and one reported the factor exactly. With inhibitory reversal zero, the sensors entered the conductance denominator and could support successive divisions [24, 37]. Separate nested, flat and locally cancellable gain tasks tested the role of factor grouping and input access (Methods; Supplementary Section S5).

The original 180-epoch comparison favored aligned serial D3 models and showed a large exact-path advantage within the LocalCA training recipe (Fig. 6A). However, all fifty D3 fits across the five credit/optimization conditions reached that cap with validation loss still falling. We therefore repeated those fifty fits and ten exact-BP D1 references with a maximum of 600 epochs, preserving the original seeds, learning rates, parameter groups and patience of thirty epochs. Both optimization recipes used Adam. These were restarts, because the historical checkpoints lacked optimizer state; source-concordance differences are retained in SI. Comparisons between budgets use validation-selected states from these same new trajectories (Fig. 6B).

Within those restarts, the LocalCA exact-path-minus-shared-soma accuracy gap changed from +10.86 points at 180 epochs to -1.52 at 600 (95% paired interval, -1.83 to -1.23), reversing in all ten seeds. Under the BP recipe, the 600-epoch exact-minus-broadcast gap was only 0.48 points (0.40–0.55; Fig. 6C). In a secondary comparison, exact-path LocalCA retained lower mean held-out cross-entropy than shared-soma LocalCA, 0.254 versus 0.276. Accuracy counts threshold crossings, whereas cross-entropy also reflects predicted probabilities. Thus the credit comparison depends on the performance metric as well as the training budget.

The forward-depth benefit persisted: exact-BP D3 exceeded D1 by 38.17 points at the longer budget. All fifty D3 fits still reached 600 epochs with declining validation loss, so this remains a finite-budget comparison. Sensor shuffling, reversed placement, grouped-point and task-family controls delimit the serial computational advantage (Supplementary Figs. S15, S18, S28 and S31). Same-seed computational reruns reproduce the original-budget comparisons (Supplementary Fig. S26). Serial structure can aid the computation while a shared somatic signal supports high classification accuracy; the earlier eleven-point gap does not establish a lasting need for exact path credit.

Reconstructed arbors provide spatial routes beyond a shared broadcast

MICrONS links reconstructed anatomy and connectivity to visual responses [38]. We used observed skeletons, soma positions and mapped contact locations and areas to model conductances, voltages and gradient fields, asking whether measured geometry supplies useful delivery patterns. This tests anatomical capacity rather than observed teaching signals.

Each inhibitory-bearing segment supplied a candidate route over its input-bearing descendants (Fig. 7A). For a site-by-route matrix A_K , diagonal excitatory-area weights W and weighted least-squares projector $P_{A_K, W}$, capture of a field $\mathbf{t}$ was

$$\mathcal{C}(A_K; \mathbf{t}) = \frac{\|P_{A_K, W} \mathbf{t}\|_W^2}{\|\mathbf{t}\|_W^2}, \quad \|\mathbf{v}\|_W^2 = \mathbf{v}^T W \mathbf{v}. \quad (7)$$

For an ensemble of perturbations, numerator and denominator sum over field columns. A common-constrained SVD dictionary provided a ceiling at the same column budget. Controls used random anatomical routes, cable-distance bins, site-shuffled routes preserving nonzero counts, and surrogate trees preserving degree and depth while changing ancestry.

The fields were gradient changes induced by small focal shunts in reciprocal cables, independently of dictionary selection. In the original eight cells, broadcast captured 31.4% of field energy, near the 32.6% rank-one ceiling but leaving most energy outside it. Subtree routes lacked a constant component whereas depth bins could broadcast, confounding common and spatial capacity. We therefore gave every family one budgeted broadcast column and $K - 1$ spatial profiles, measuring total and residual-field capture and recording actual rank and nonzero count. The protocol was fixed before extension to 47 disjoint cells from the same mouse and a reconstructed second mouse (Fig. 7).

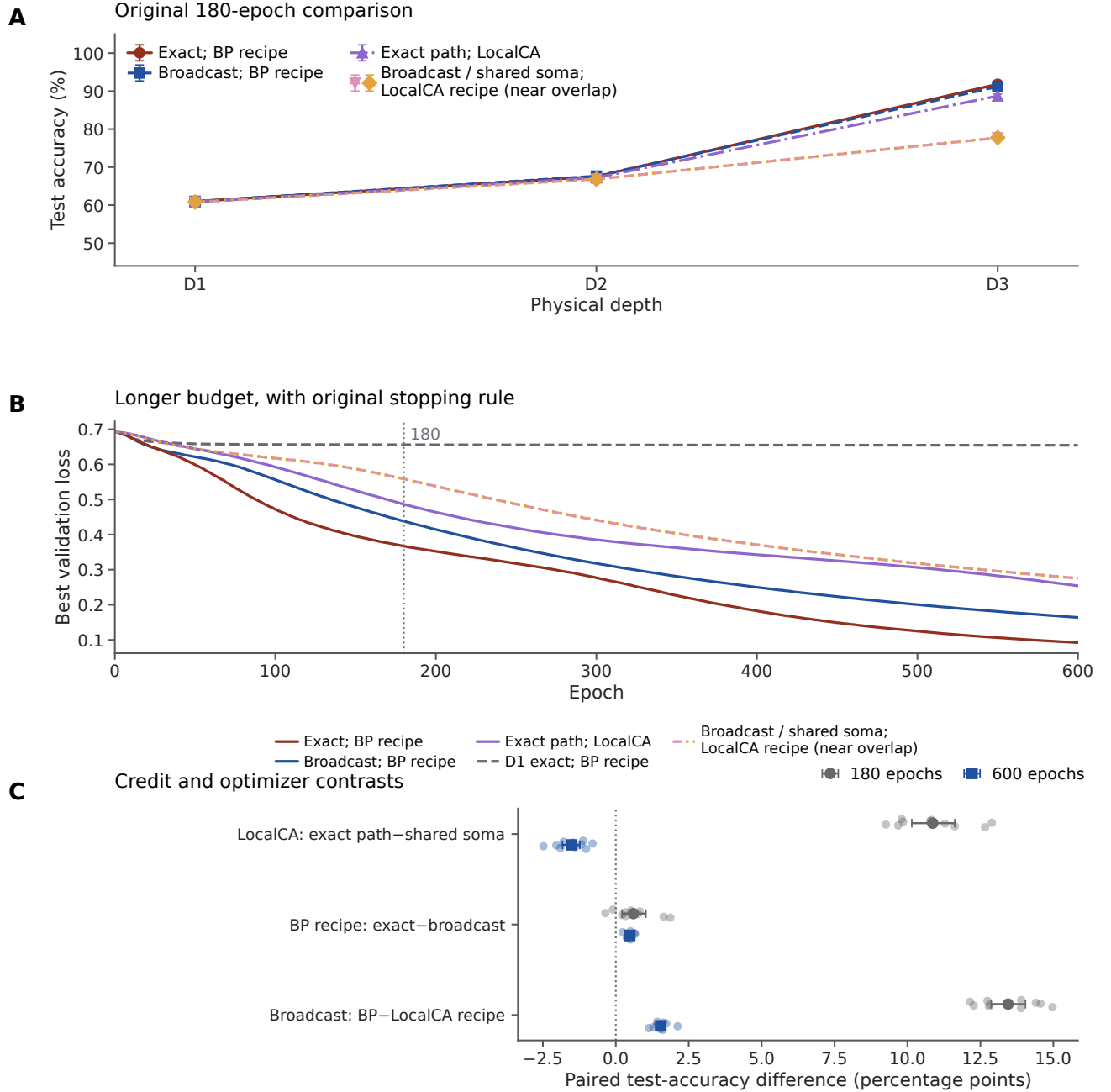


Figure 6: Physical depth and credit delivery under two training budgets. **A**, Original 180-epoch comparison with fixed branch resources and five delivery/optimization conditions; means and 95% seed-bootstrap intervals. Broadcast and shared-soma LocalCA nearly overlap; both curves have a paired legend entry. **B**, Best validation loss in sixty new restarts with maximum 600 epochs and the original patience thirty. Stopped runs retain their best loss, so each curve includes the same ten seeds throughout. Colored curves are D3; gray dashed line is exact-BP D1. The vertical line marks 180 epochs. Broadcast autograd under the LocalCA recipe and shared-soma LocalCA nearly overlap; both are retained with a paired legend entry. **C**, Paired test-accuracy contrasts between validation-selected states within the 180- and 600-epoch windows of those same trajectories. Faint points are ten paired seed differences; larger symbols show means and descriptive 95% whole-seed bootstrap intervals. Both recipes use Adam with their original parameter groups, learning rates and update options. Exact-path LocalCA has lower loss but lower accuracy than shared-soma LocalCA at 600 epochs. All outcomes are retained; the larger budget does not establish convergence.

At eight columns in the disjoint 47-cell cohort, broadcast-plus-subtree routes captured 0.619 of total energy, compared with 0.511 for depth bins and 0.436 for the site-shuffled control (Fig. 7B). The paired ancestry-minus-depth advantage was 0.108 (95% cell-bootstrap interval, 0.077–0.135), positive in 45 of 47 cells. After removing broadcast, residual-energy capture was 0.527 for ancestry and 0.392 for depth bins (Fig. 7C). Degree–depth surrogates captured 0.578 of total energy; the smaller ancestry advantage over this control was 0.041 (0.020–0.062). Thus coarse geometry accounts for much of the capacity, with additional information in the retained ancestry.

The original eight cells retained the same ordering (Fig. 7D). In the Pinky second mouse, eight of ten qualifying cells supported the eight-column comparison. Their 9–13 input-bearing sites put ancestry near the residual-capture ceiling, so lower-budget comparisons and exclusions remain separate from the larger arbors. Earlier route-generated positive controls test imposed alignment, distinct from these independent fields (Supplementary Figs. S10, S20 and S27).

Direct E/I labels covered 4.71% of disjoint-cohort inputs; the original cohort’s 73.0% direct-versus-proxy agreement concerned jointly labeled contacts. Nonzero site-by-route coefficients are a delivery-cost proxy, not feedback synapses or metabolic expenditure (Fig. 7E). Geometry, labeling and compression sensitivities are in Methods and Supplementary Figs. S12, S25 and S33.

Focal shunting sets gains on ancestry-defined credit routes

Morphology supplies spatial routes; conductance can regulate their amplitudes. We examined this connection by comparing a focal inhibitory shunt with a current injection matching its baseline focal current in reconstructed reciprocal cables. A separately solved somatic current restored baseline somatic voltage after either intervention. The endpoint was the change in the modeled excitatory-conductance gradient.

Let G be the passive conductance matrix and $\mathbf{b}$ the current-source vector, so that baseline voltages satisfy $G\mathbf{V} = \mathbf{b}$. With $\mathbf{e}_k$ the unit vector at compartment k , adding inhibitory conductance κ_k at compartment k changes both G and its source: $G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top$, $\mathbf{b}' = \mathbf{b} + \kappa_k E_1 \mathbf{e}_k$. The matched injection supplies baseline current $\kappa_k (E_1 - V_k)$ without changing G . The adjoint $\mathbf{q}$ gives loss sensitivity to current injected at each compartment. For a fixed somatic loss derivative, it changes by

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (8)$$

The column $G^{-1} \mathbf{e}_k$ gives the voltage response to unit current at k . On a passive tree with a somatic loss, the resulting adjoint change has an exact ancestry structure: all descendants of the shunt share the attenuation factor $[1 + \kappa_k (G^{-1})_{kk}]^{-1}$, while each sister block along the path to the soma shares its own factor (Fig. 8A; SI S4). These disjoint blocks form an ancestry partition. Weighting their indicators by the baseline soma-to-site transfer gives a dictionary on which the shunt acts through diagonal gains, providing a physical instance of a diagonal channel-gain matrix, denoted Γ_u . The diagonal form applies in this partition basis; it need not hold for an arbitrary selected set of overlapping routes.

We compared descendants $D(k)$ with off-route sites $C(k)$ at similar depth. A separate kernel criterion tests whether the adjoint change is larger below the shunt (Methods); within-cell, reversal-potential and contact-label controls are in Supplementary Fig. S21. The analyzed gradient was sensitivity to open excitatory conductance, $\gamma_i = q_i (E_E - V_i)$. The local driving force $E_E - V_i$ can change differently within each block, so the complete synaptic gradient need not share a single block gain. With primes denoting post-intervention values and ε_γ a numerical floor defined in Methods, we measured absolute log change, $m_i = |\log[(|\gamma'_i| + \varepsilon_\gamma)/(|\gamma_i| + \varepsilon_\gamma)]|$, and localization $\Lambda_k = \text{median}_{D(k)} m_i - \text{median}_{C(k)} m_i$. Positive localization means a larger gradient change below the shunt.

In the permissive normalized model, shunts changed descendant gradients more strongly than depth-matched off-route gradients, with increasing localization at higher dose (Fig. 8B; Supplementary Fig. S48). Current injection changed descendant gradients roughly threefold less. Its off-route changes were zero by construction under the soma-restoring control; this is not a general claim that current injection cannot change a gradient. Holding the post-shunt voltage but replacing its adjoint with the baseline adjoint removed most localization, supporting a transport contribution beyond the voltage change alone (Fig. 8C).

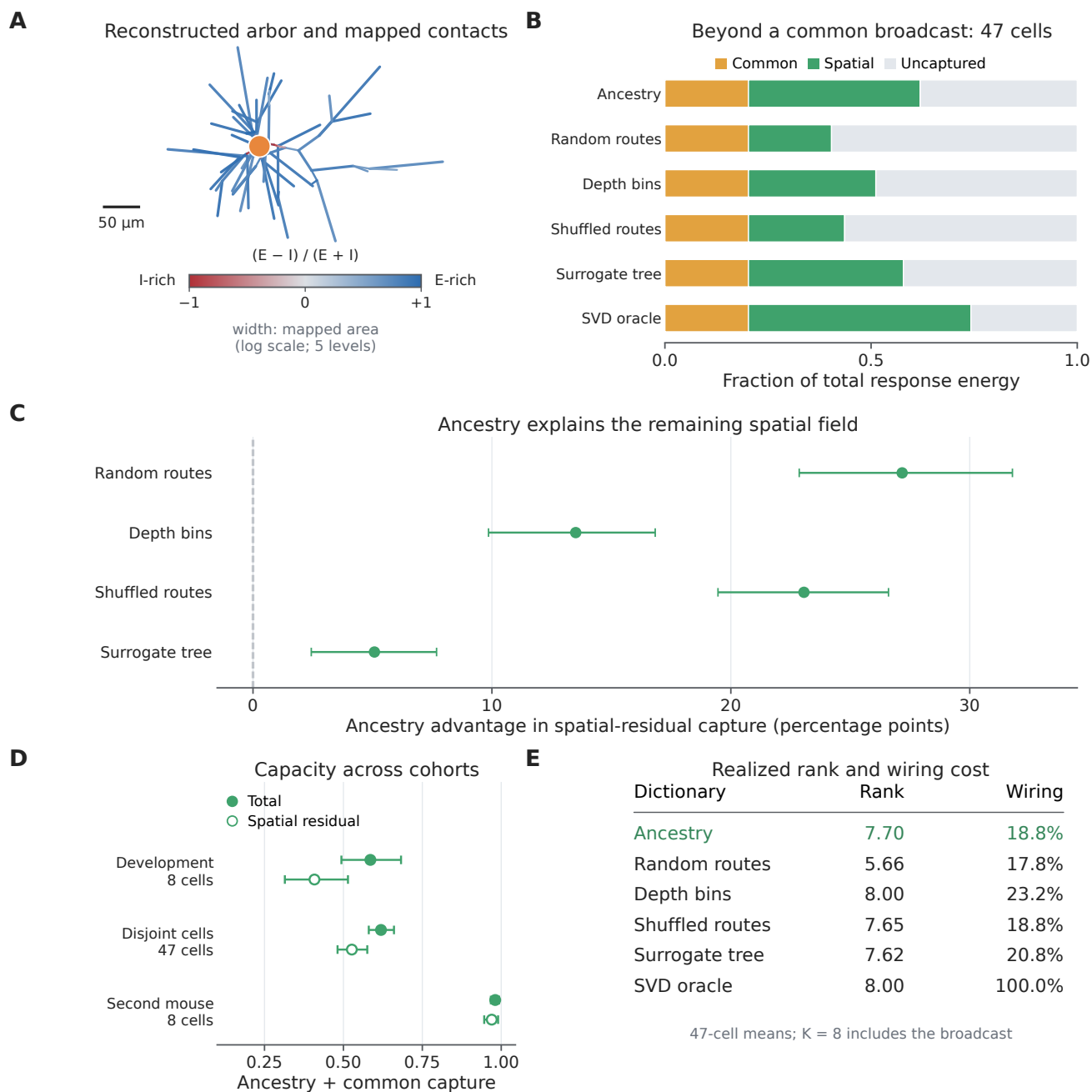


Figure 7: Ancestry captures spatial response structure beyond a shared broadcast. **A**, Median-sized reconstructed arbor from the initial cohort. Color indicates mapped E/I area balance; width indicates total contact area; scale bar, 50 μm . **B**, Mean energy decomposition for modeled focal-shunt responses in 47 disjoint v661 cells from the same mouse. Every dictionary has one common broadcast and seven spatial columns. **C**, Paired ancestry advantages in residual energy capture after removing broadcast. Random routes sample anatomical sites; shuffling preserves column nonzero counts; surrogates preserve segment depths and parent out-degrees. **D**, Separate original, v661 and Pinky cohorts. Filled/open symbols indicate total/residual capture; Pinky's eight eligible cells contain only 9–13 excitatory-bearing sites each. **E**, Actual rank and nonzero density relative to dense eight-column wiring in v661. Equal column count does not ensure equal rank or wiring; ancestry and shuffling have identical nonzero counts. Randomized controls average 200 draws per cell. Symbols and bars show cell means and 95% bootstrap intervals; these describe within-cohort consistency. Capture uses squared excitatory-area-weighted field energy. The common-constrained SVD is an oracle capacity ceiling, and modeled fields are not observed teaching signals.

Physical calibration constrained the spatial selectivity of the gradient change. At axial resistivity $R_a = 150 \Omega \text{cm}$ and specific membrane resistance $R_m = 15,000 \Omega \text{cm}^2$, mean shunt-minus-current localization was small and opposite-signed in the initial eight cells and a disjoint 45-cell cohort from the same mouse: -0.0019 and $+0.0024$, respectively. Signed changes showed attenuation of both descendant and off-route gradients at high resistance; lowering resistance primarily increased descendant attenuation (Fig. 8D). This increased the localization contrast by more than an order of magnitude (Fig. 8E). Distributed background conductance produced the same qualitative recovery at fixed high membrane resistance (Supplementary Fig. S11), consistent with a dependence on electrotonic state [25, 39].

Adding weak voltage-dependent channel currents produced a dose response close to the passive response at $R_m = 1,000 \Omega \text{cm}^2$ with one baseline-leak unit of distributed background conductance (SI S9). This checks the local linearization around solved steady states. The baseline Jacobian included gate derivatives but remained fixed across doses, so the check does not establish robustness to strongly active dendrites or channel kinetics.

Together, the exact partition identity and the electrical calibration connect the morphology-defined dictionary to conductance-dependent route gains [26]. Their spatial selectivity depends on cable state. These imposed, state-matched perturbations measure modeled transport, rather than endogenous error production, observed plasticity or a learning advantage.

Measured responses test ancestry alignment and expose a transfer-geometry limit

Measured responses ask whether functional relationships preferentially follow ancestry. Seven MICrONS targets had complete arbors and imaged partners: 69 partners in the selected-scan dataset and thirteen eligible scans in the scan-complete analysis. These are recordings during stimulus presentation, not learning.

The direct structure–function test asked whether partners with more shared dendritic ancestry had more similar condition-mean responses, after controlling Euclidean separation and soma-to-contact path-length differences. Mean partial rank correlations were -0.069 for selected scans and -0.048 for all eligible scans; both target-bootstrap intervals included zero (Fig. 9A). This empirical test provides no support for preferential ancestry alignment in the sampled partners. Its scale differs from population-level like-to-like wiring [40].

A separate offline learning comparison used presynaptic responses to predict each target’s response through its complete passive tree. Trainable nonnegative excitatory conductances received input proxies at the partners’ dominant mapped contact segments; a linear somatic readout predicted the target. Stimulus identities were held out, and transformations and model choices used training data only. Exact learning supplied trial-specific spatial sensitivities. Each restricted rule instead projected the baseline soma-to-site transfer profile $\mathbf{v}_0$ into a four-route dictionary D , giving $\mathbf{v}_D = P_D \mathbf{v}_0$, and delivered

$$\hat{\delta}_{i,D}^V = \delta_i^{\text{out}} \mathbf{v}_D. \quad (9)$$

Here δ_i^{out} is the scalar output error on trial i . The readout and bias always received exact gradients; only input-conductance credit differed.

The realized dictionaries explain what this comparison can test. Selected subtree routes were predominantly distal supports: across scans, they reached 48.0% of mapped partner inputs and contained an average of 1.31 inputs per route. Several inputs can share one physical segment. Six of thirteen scans had only single-input routes (Fig. 9B,C). Moreover, one unrestricted fixed baseline-transfer profile already reconstructed 98.4% of the exact field and 97.6% of exact updates. The model therefore offered little trial-dependent spatial variation for a richer encoder to exploit. Its learning comparison chiefly diagnoses loss of a nearly fixed transfer profile when that profile is restricted to incompletely covering supports.

Consistent with that geometry, the subtree rule had no reliable held-out advantage over random anatomical routes or site shuffles. Exact learning performed best, while ridge regression captured most of its predictability (Fig. 9D). Exact learning improved normalized MSE over ridge by 0.015 on average (95% target-bootstrap interval, 0.004–0.027). Here NMSE is relative to predicting the training mean, unlike the known population-variance normalization in the artificial tasks. Ten split/initialization replicates were averaged within scans and then within targets; the seven targets were the inferential units.

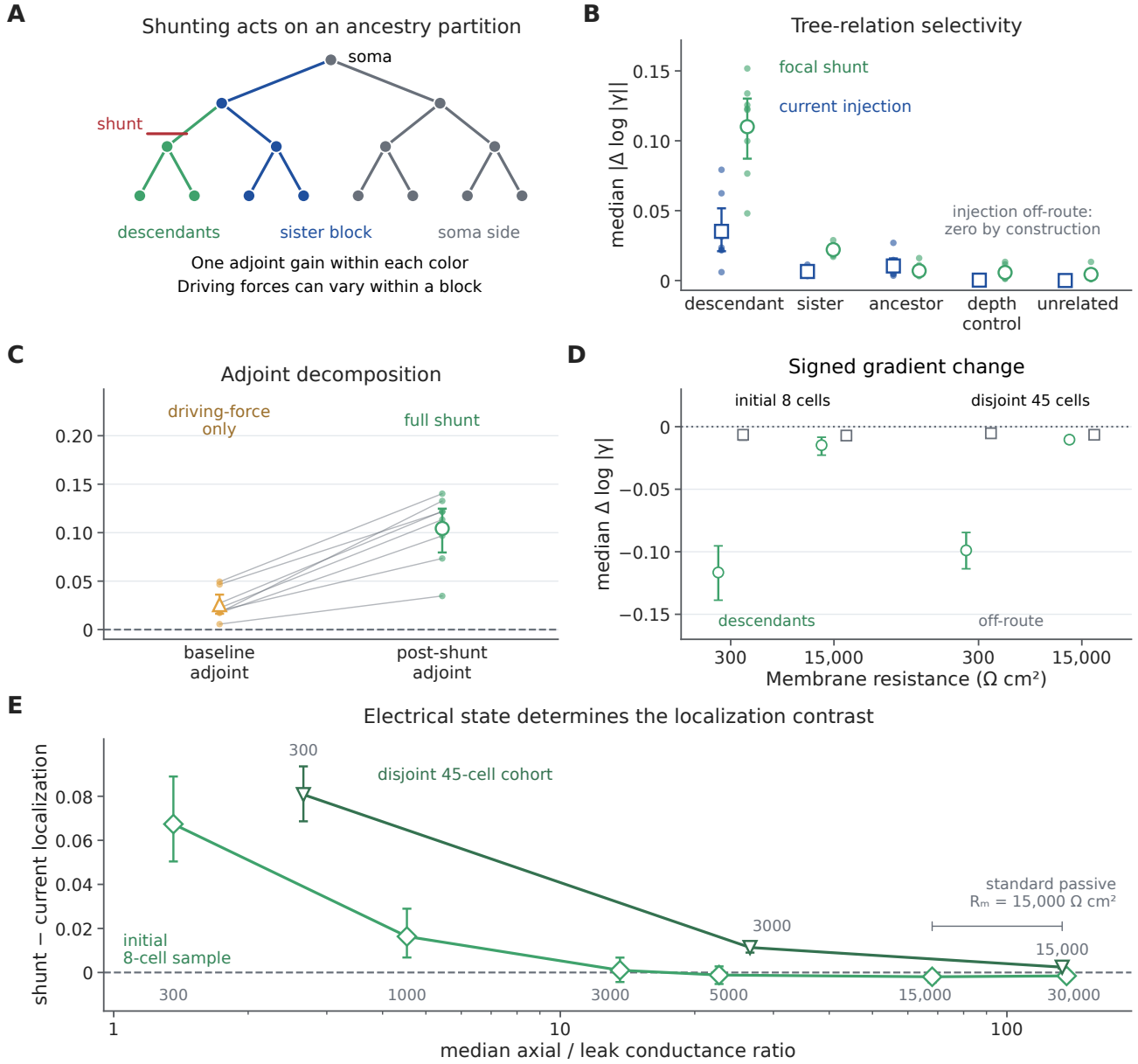


Figure 8: Focal shunts set ancestry-defined gains whose selectivity depends on electrical state. **A**, On a passive tree with a fixed somatic loss derivative, a shunt gives the same adjoint attenuation factor to every site in each colored ancestry block. The resulting gain is diagonal in the baseline-weighted partition dictionary; local synaptic driving forces can still vary within a block. **B**, Absolute log change in excitatory-conductance gradient magnitude by tree relation at unit dose, in eight reconstructed cells under permissive normalized passive parameters. A current injection matches baseline focal current, and a separately solved somatic current restores baseline somatic voltage after either intervention. Off-route injection zeros arise from this construction. **C**, Keeping the post-shunt voltage while restoring the baseline adjoint removes most localization. Lines pair cells. **D**, Signed log change in gradient magnitude at the two physical-calibration endpoints, with $R_a = 150 \Omega \text{ cm}$ and $R_m = 300$ or $15,000 \Omega \text{ cm}^2$. Circles and squares show descendants and depth-matched off-route sites; negative values denote attenuation. Site medians are averaged within cells; symbols and whiskers are cohort means and 95% whole-cell bootstrap intervals (20,000 draws). **E**, Electrical calibration in the initial eight cells and a disjoint 45-cell cohort from the same mouse. The horizontal axis is the cohort-median axial/leak ratio at fixed axial resistivity $R_a = 150 \Omega \text{ cm}$; numerical labels give membrane resistance R_m in $\Omega \text{ cm}^2$. Shunt-minus-current contrasts are small and opposite-signed at $R_m = 15,000$, and increase at lower resistance. Symbols and whiskers are cohort means and 95% cell-bootstrap intervals after averaging sites within cell. These state-matched perturbations measure modeled transport, not plasticity or learning benefit.

434 Allowing optimal trialwise coefficients within the same dictionaries barely improved ancestry update reconstruc-
435 tion (Fig. 9E). For trial-by-parameter update matrices U , we measured

$$\mathcal{M}_\Delta(D) = 1 - \frac{\|U_{\text{exact}} - U_D\|_F^2}{\|U_{\text{exact}}\|_F^2}, \quad (10)$$

436 where the squared Frobenius norm sums squared entries. Ancestry rose from 0.232 for its fixed profile to 0.244 with
437 oracle coefficients, compared with 0.385 for random routes and 0.468 for shuffles. Thus the primary limitation was
438 the selected span’s geometry, not coefficient freezing. The re-executed checkpoint discrepancies and exact oracle
439 definitions remain in Methods and Supplementary Table S9.

440 Input-removal and noise controls test dependence on recorded partners (Supplementary Fig. S34); imposed-
441 alignment fields establish learning capacity when targets lie in the anatomical span (Supplementary Fig. S5). Neither
442 tests endogenous teaching. The empirical ancestry-similarity test and modeled transfer diagnostic therefore bound
443 distinct biological claims.

444 Discussion

445 Dendritic trees provide spatial addresses through which learning can distinguish conflicting synaptic demands. Which
446 distinctions are useful depends on the task and the local sensitivity already available to the learner. Morphology
447 supplies delivery patterns, teaching mechanisms supply their coefficients, and conductance regulates their gains.
448 Image learning benefits mainly from preserving neuronal identity, including with random between-neuron feedback.
449 Contextual conflict favors branch selection, supplied through feedback or local eligibility gating. Hierarchical
450 distractors favor ancestry at a bandwidth predicted by the generator’s coefficients. These results distinguish useful
451 information from its physical carrier.

452 The interaction comparison shows why input rank alone is insufficient. Pairwise and quartic targets share input-
453 sensitivity spectra and a compatible tree but develop different credit spectra and tolerance of fixed profiles, despite
454 initial calibration and common-rate controls. Spatial direction matters alongside dimension: even a nearly one-
455 dimensional field can be poorly captured by uniform broadcast. These are learned-state properties, not retrospective
456 support for the failed initialization selector. Intermediate dictionaries, adaptive profiles and alternative dynamics
457 remain plausible solutions.

458 Conductance trees connect this distinction to a dendritic computation. Simple credit learns some compatible
459 teachers accurately, whereas opposing branch tuning under contextual inhibition benefits strongly from routed credit;
460 three supplied spatial patterns with oracle coefficients suffice in the tested circuit. Forward compatibility remains
461 essential: the scalar multi-affine cut criterion connects input grouping to hierarchical tensor descriptions and supports
462 structure estimation from noisy examples [41–43]. Serial composition likewise helps remove distributed nuisance
463 factors, but longer training reverses the LocalCA accuracy advantage of exact credit while preserving its lower
464 cross-entropy. Task rank, physical depth and credit resolution therefore describe different constraints.

465 Reconstructed arbors provide ancestry capacity beyond a common broadcast, including in the disjoint 47-cell
466 cohort; degree–depth surrogates explain much of that capacity. Focal shunting supplies a biophysical link by
467 reweighting the baseline-weighted ancestry partition, with additional local changes from synaptic driving forces.
468 Cable state determines their spatial selectivity. This concerns sensitivity transport, distinct from inhibition that
469 computes errors or regulates forward gain and load [44–47].

470 Measured ancestry–response similarity showed no preferential alignment in the small recorded cohort. The
471 offline learning comparison chiefly tests preservation of an almost fixed transfer profile and offers limited evidence
472 about endogenous route use. A direct experiment would compare plasticity below a focal inhibitory site with
473 electrotonically matched off-route sites while matching somatic voltage, first-order current and outcome signal.
474 Changing conductance state would test the predicted separation between shared and ancestry-selective effects
475 [22, 48, 49].

476 The framework complements somatic prediction, segregated teaching, interneuron-mediated errors and burst
477 plasticity [28, 30–32], alongside cable-learning and engineered gating models [35, 50–53]. Spikes, temporal

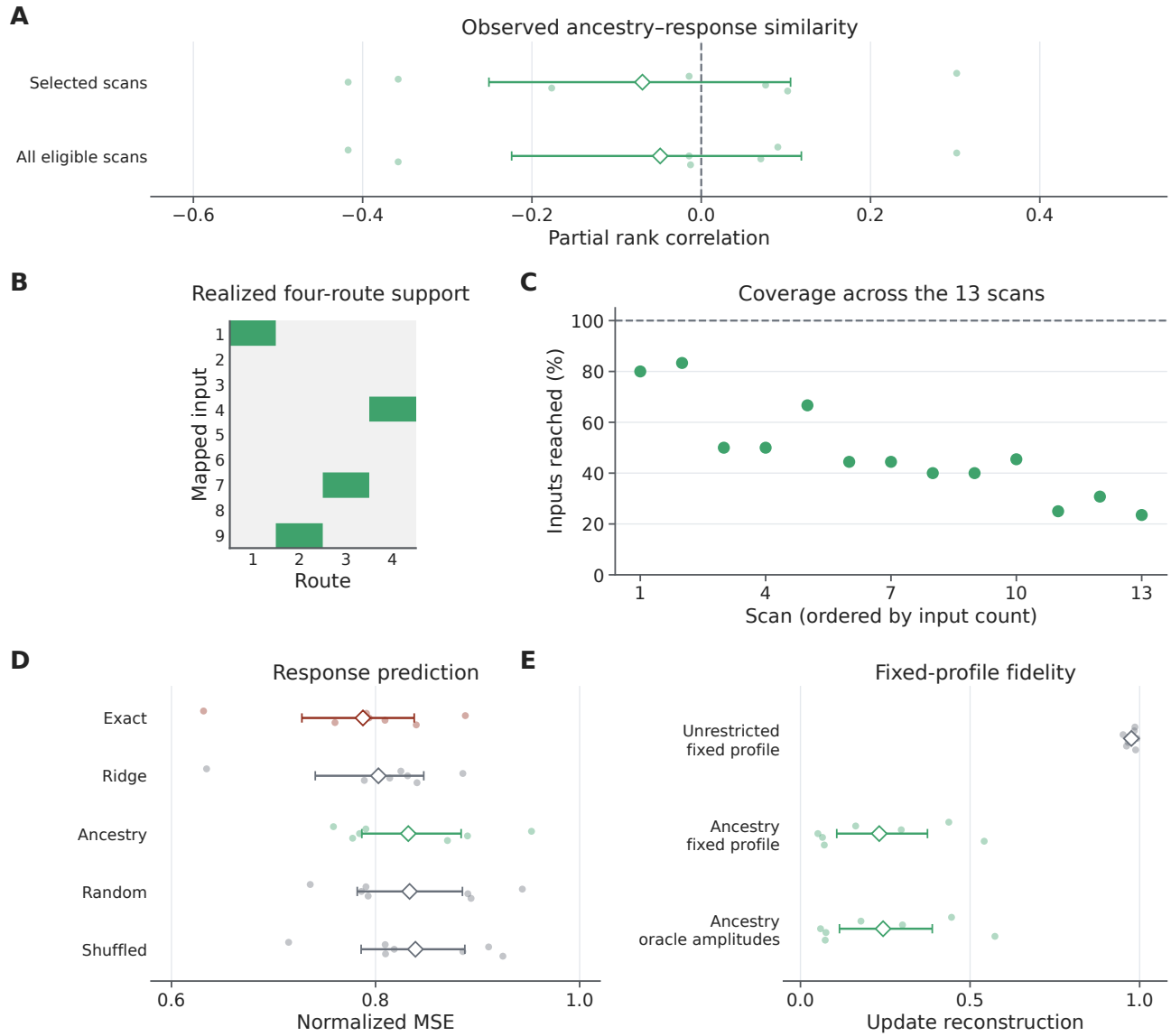


Figure 9: Observed ancestry alignment and a separate transfer-geometry diagnostic. **A**, Target-level partial rank correlations between shared ancestry and response similarity after controlling Euclidean separation and contact-depth differences. Selected scans contain 69 partners; all eligible scans comprise thirteen scans of the same seven targets. **B**, Actual input-by-route support in the scan with median mapped-input count, chosen without response outcomes; green entries indicate nonzero routes. Input-segment and route identifiers are retained in Source Data; several inputs can share a segment. **C**, Fraction of mapped inputs reached by selected ancestry routes in each scan, ordered by input count; scans are descriptive repeated observations within targets. **D**, Held-out normalized MSE for exact and restricted nonlinear learning and training-tuned ridge regression. NMSE uses the training-mean predictor as baseline. **E**, Common-checkpoint update reconstruction by the unrestricted fixed transfer profile, its restriction to ancestry routes, and optimal trialwise amplitudes within that ancestry span. A value of one is exact reconstruction. The last two differ in coefficient access, not in spatial support. In **A,D,E**, dots are seven target values; diamonds and whiskers show means and 95% target-bootstrap intervals. Panels **B–E** describe an offline model comparison and do not measure biological learning signals.

eligibility, recurrence and channel kinetics may change the available profiles and coefficient estimation [54, 55]. The organizing question remains which task-relevant distinctions the arbor and learning mechanism can jointly preserve.

Methods

The Methods define the models, task manipulations and inferential units behind the main comparisons. Supplementary Sections S1–S4 provide derivations and the interaction studies; S5 gives complete artificial-network protocols; S6–S12 cover anatomy, perturbations and measured responses. The supplementary cohort index lists all experiments. Resolved configurations, seeds, checkpoints, exclusions and source hashes accompany the numerical tables. Reusing a cohort for a diagnostic does not add independent observations.

Conductance models and delivery coordinates

In a directed tree, $[b_1, \dots, b_D]$ lists branching factors from soma to tips, giving $\sum_{d=1}^D \prod_{r=1}^d b_r$ nonsomatic compartments. A child c supplies output $a_c = f_c(V_c)$. With leak g_n^L , aggregate open excitatory/inhibitory conductances $G_n^X = \sum_{i \in \mathcal{J}_n^X} g_i^X x_i^X$ for contact class X , and child couplings $g_{c \rightarrow n}^{\text{den}}$, the steady voltage is

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + G_n^E E_E + G_n^I E_I + \sum_c g_{c \rightarrow n}^{\text{den}} a_c \right), \quad (R_n^{\text{tot}})^{-1} = g_n^L + G_n^E + G_n^I + \sum_c g_{c \rightarrow n}^{\text{den}}. \quad (11)$$

Here $x_i \geq 0$ is contact activity, $g_i \geq 0$ its conductance, and E_X its reversal potential. A terminal-to-soma pass evaluates the directed steady state. Artificial conductance trees use dimensionless leak one and reversals $E_L = E_I = 0$, $E_E = 1$. Monotone parameter transforms enforce nonnegative conductances. Branch-output nonlinearities, termed reactivations in the implementation, are identity functions when disabled. Both excitatory and inhibitory contacts learn in the regular-tree experiments.

For a reciprocal cable residual $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = 0$, the nonsingular voltage Jacobian J_V gives the adjoint

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}, \quad \partial \mathcal{L} / \partial g_i = -\mathbf{q}^T \partial \mathbf{F} / \partial g_i. \quad (12)$$

With passive current balance, the synaptic gradient is $x_i(E_i^{\text{rev}} - V_n)q_n$. This established sensitivity calculation supplies the reference for restricted credit [9, 10, 29, 35]. Directed LocalCA combines local eligibility with delivered credit. For output $a_n = f_n(V_n)$, activation error $\delta_n^a = \partial \mathcal{L} / \partial a_n$ gives voltage error $\delta_n^V = f'_n(V_n) \delta_n^a$. The image implementation retains f'_n in local eligibility and replaces δ_n^a ; its gate updates also retain their exact local derivatives. Thus a fixed activation-space profile can correspond to a state-dependent voltage-space profile. Supplying the exact field recovers the exact selected gradients.

The raw-additive control retains the graph but sums signed inputs without the conductance denominator or reversal-potential driving force. Historical normalized-additive controls additionally standardize voltage across stage coordinates within each example; Supplementary Figs. S1–S4 identify those variants. The optional four- and five-factor rules add stage-level preconditioners and are described in Supplementary Section S2.

Writing a delivery dictionary as $A\Gamma$ separates spatial profiles in A from diagonal channel gains in Γ . Nonzero gains preserve the column span and oracle capture, while changing delivered coefficients and conditioning. A passive focal shunt admits this representation in its baseline-weighted ancestry partition; arbitrary overlapping dictionaries need not. Physical stage count, feedback rank, external error dimension and state-dependent path resolution are distinct resources (Supplementary Tables S13–S14).

Image feedback ladders and checkpoint diagnostics

For example i , the trained linear decoder supplies activation errors $\delta_{i,u} = \partial \mathcal{L} / \partial a_{i,0,u}$ at the somatic outputs of neurons u . The strict scalar is their mean unless its magnitude is at most 10^{-6} , in which case it is the largest-magnitude coordinate, preserving that coordinate’s sign. The same scalar reaches every core compartment, including the somatic gate. Neuron-specific feedback repeats each neuron’s own coordinate throughout its arbor; exact transport

multiplies it by the local path derivatives. The historical matched-width fallback retained neuronal identity at the 128-wide somatic stage, affecting 640 of 2,413,312 core parameters. A separately trained strict-scalar control tests that exception (Supplementary Table S16).

The new MNIST ladder uses 128-soma [3, 3] shunting and raw-additive networks, with 40 excitatory and 20 inhibitory contacts per nonsomatic compartment. Within each architecture, rules share initial contact construction, initialization policy, decoder and data streams. Pixels are divided by 255 and flattened without further normalization or augmentation. Each seed divides the 60,000 training images into 48,000 training and 12,000 validation examples; all 10,000 official test images remain held out. Training uses 180 epochs, batches of 256 and the minimum-validation-cross-entropy checkpoint, without early stopping. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ and no weight decay. Original rates are 0.0015 for input synapses and decoder, 0.0007 for branch couplings and 0.0005 for reactivation gates.

Projected $K = 1$ averages all twelve nonsomatic activation errors. The $K = 3$ dictionary instead averages each proximal compartment with its three distal children. Both use the exact field and preserve its mean at identical weights; their norms can differ. Both projected rules retain exact somatic errors and local activation derivatives. A decoder-only reference trains the readout on the initialized core, whose unchanged hash is verified. Three development seeds compare rate multipliers 0.3, 1 and 3 for every rule and architecture, giving 108 fits. Validation alone selects rates. Ten fresh paired seeds per architecture and rule give 120 selected-rate and 120 original-common-rate outcomes from 190 unique fits; 50 fits serve both policies. The projected- $K = 1$ addendum was frozen before fresh outcomes. Complete protocols, clipping and excluded implementation checks are in Supplementary Section S5.

The original fifteen-seed MNIST ladder remains in Supplementary Fig. S45. Fashion-MNIST uses ten paired seeds; the raw-additive CIFAR-10 ladder uses twenty and a 20-soma [3, 3, 3, 3] tree. Their distinct protocols and initialization are retained in SI. The separate fifteen-seed direct-feedback-alignment factorial replaces the decoder derivative by a fixed Gaussian output-to-neuron matrix, scaled by inverse square root of output dimension. It pairs random-feedback fits with the original exact-readout reference. Ownership controls permute neuronal recipients without changing coordinate values or bandwidth.

Common-checkpoint diagnostics compare rules at identical weights and states. Exact path transport is checked against automatic differentiation. New capture measurements use 2,048 held-out images at initial and exact-trained states; activation- and voltage-space metrics are kept separate. Gradient cosine, error magnitude by depth and mean-subtracted path energy are distinct summaries; none alone certifies learning performance. Historical depth and noise cohorts, including their task-identity and conductance-validity records, are specified in Supplementary Section S5.

Contextual conflict and ancestry routing

In the branch-conflict task, one context-selected branch determines a logistic output while all B branches form ungated local eligibilities. Nonselected Fashion-MNIST views are replaced independently by opposite-class images with probability χ . Exact feedback is $\delta_i \mathbf{1}[b = b_i^*]$; shared feedback is δ_i/B , where δ_i is output error and b_i^* the selected branch. Twenty paired seeds cross $B = 2, 4, 8$ with eight conflict doses. The algebraically equivalent gated-point calculation and BP receive the same updates; cyclic derangement preserves rank and support while changing the recipient.

Under centered, symmetric class evidence, shared-mode compatibility is $s(\chi) = 1 - 2\chi(B - 1)/B$, with zero at $\chi_c = B/[2(B - 1)]$. This mean-field prediction is separate from a trained accuracy crossing. The latter is reported both by interpolation of the mean curve and by each seed's first sampled at-or-below-chance dose, retaining non-crossings. Checkpoint re-execution reproduces endpoints and compares exact/shared fields at both common and own trained states (Supplementary Fig. S29).

The ancestry factorial contains eight eight-dimensional blocks. Context c_i and binary label sign s_i are balanced, and inputs are $\mathbf{x}_{i,b} = a(b, c_i)s_i \mathbf{t}_b + \boldsymbol{\epsilon}_{i,b}$, with seed-specific unit teachers $\mathbf{t}_b$ and Gaussian noise $\boldsymbol{\epsilon}_{i,b} \sim \mathcal{N}(0, 1.1^2 I_8)$. The selected coefficient is one; sibling, same-half nonsibling and opposite-half coefficients are $-0.15, -0.45, -0.75$. A leaf permutation changes task-to-route alignment without changing the streams. For budget K , ancestry feedback has context-by-block entries $\Phi_{cb}^{(K)} = \sqrt{K/8} \mathbf{1}[g_K(\pi(c)) = g_K(b)]$, where g_K indexes contiguous groups and π maps

streams to leaves. Thus every row has unit Euclidean norm.

The update is $\Delta \mathbf{w}_b = -(8\eta/N) \sum_i \delta_i \Phi_{c,b}^{(K)} \mathbf{x}_{i,\pi^{-1}(b)}$, with N training examples and rate η . Each seed supplies 2,048 training, 2,048 test and 1,024 switch examples. Full-batch descent uses rate 0.3, eighty training epochs and twenty switch epochs. Controls include deranged ancestry, depth-interleaved and random-sparse supports, dense rank- K fields and a training-informed PCA reference. The best matched control is the per-seed post-training maximum over the four non-PCA controls. Integer test counts preserve ties in the named Wilcoxon comparison; Holm correction covers four budgets. Alternative mean-sign-flip and historical BH analyses remain explicitly labeled in Supplementary Table S2. The separate noisy-context encoder and cue-delay experiments are fully specified in Supplementary Section S5.

Update utility and prospective selection

Let the exact gradient be $\boldsymbol{\mu}$ and the delivered update estimator have mean $\mathbf{m}$ and covariance Σ_{route} at the current weights. For an L -smooth loss and a nonnegative step η , expected loss after the update is bounded above by $\mathcal{L} - \eta \boldsymbol{\mu}^\top \mathbf{m} + (L\eta^2/2)(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})$. Optimizing this expression gives utility $U = [\boldsymbol{\mu}^\top \mathbf{m}]_+^2 / [2L(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})]$, with zero for a zero denominator. Utility certifies one-step progress under the stated moments; it is not a long-horizon ordering.

Synthetic quadratic screens isolate signal retention, noise rejection and fixed gains over sixteen Haar coordinates, without a forward neuron (Supplementary Fig. S46). Their matched signal/noise hierarchy is a designed crossover. The trained factorial’s retrospective utility uses actual full-batch updates and a global logistic-curvature bound; choosing maximum bandwidth matches its selection success. The separate prospective linear experiment retains all 12,800 fits, the initialization selector’s loss to fixed maximum-budget and rank-only baselines in every seed, and follow-ups forecasting evolving weights and noise. The 25,600-fit dynamics extension and observed-context-count control are reported in Supplementary Figs. S35 and S38–S39, with sealed choices, complete methods and computational costs.

Interaction compatibility, fixed profiles and learned spectra

For a subtree leaf set S , C_S arranges the target’s Fourier interaction coefficients by products inside and outside S ; C_S° removes the constant inside row. Scalar multi-affine trees represent the target exactly if and only if every centered cut has rank at most one. If $\mathcal{S}(T)$ contains all proper cuts of tree T and singular values are ordered from largest to smallest,

$$\text{MSE}(f, \hat{f}_T) \geq B(T; f) := \max_{S \in \mathcal{S}(T)} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (13)$$

Overlapping cut tails cannot be added as a lower bound. Supplementary Section S4 gives the constructive proof, minimum-depth algorithm and its relation to hierarchical tensor representations. The twelve-tree census, noisy-query estimator, end-to-end learning pipeline, conductance grouping controls and Boolean truth tables remain in Supplementary Figs. S36–S44. The theorem’s assumptions do not extend automatically to general conductance ancestors.

The matched credit experiment uses twenty fresh seed blocks after three development seeds select rates with equal tuning budgets. Matching and quartic conditions share tree, leaf assignment, initialization, inputs, additive noise and minibatch order. Exact, unit, initially calibrated and sign-only profiles use the same local eligibilities; calibration averages initial path derivatives over 256 unlabeled examples. Selected and common-rate outcomes are retained for SGD and Adam, giving 432 development and 1,200 fresh fits. Algebraic coefficients remain within $[-2, 2]$, and all bound contacts are recorded. Fits stop at 1,024 steps, with seven saved checkpoints. Noisy held-out NMSE divides by clean target variance: one for matching and nested, one-half for quartic. All have noise variance 0.0225. The primary Adam contrast subtracts the pairwise calibrated-minus-exact gap from its quartic counterpart.

For evaluation-example-by-site path matrix Q , the best fixed profile is the leading right singular vector. Its capture is $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$, an oracle reconstruction with example-dependent amplitudes. Additional diagnostics evaluate uniform, calibrated and ancestry profiles, loss-residual weighting, eligibility-weighted updates and root-mean-square

normalization of each site. Participation-ratio effective rank is $(\text{tr}C)^2/\text{tr}(C^2)$ for uncentered $C = Q^T Q/N$. At fixed weights and inputs, changing labels leaves the path field unchanged; eighty relabeling controls verify this. Independent checkpoint analysis and faithful replay of 600 earlier fits provide numerical and trajectory checks. Complete definitions, pairing audits, optimizer effects and the positive-conductance boundary are in Supplementary Section S4.

Conductance trees with context-dependent feature tuning

The seven-compartment model uses identity branch outputs in Eq. 11. Two independent $z_j \sim \mathcal{U}[-2, 2]$ generate positive inputs $\exp(\pm z_j)$, supplied to both subtrees. Every terminal has two excitatory and two inhibitory gains. Binary context inhibits one proximal compartment; teacher tuning is aligned or opposed across the two branches. All 24 positive conductances are parameterized by their logarithms. Teacher log gains receive independent Gaussian perturbations of standard deviation 0.15, and student initialization uses 0.4. Full gains, generators and exact eligibility derivatives are given in Supplementary Section S4.

Four credit rules share 2,048 training, 1,024 validation and 4,096 test examples, 128-example minibatches and the same initialization. A separate 512-example unlabeled sample fixes the calibrated profile and the three dictionary columns. Projection coefficients use the current exact field. Development comprised 288 fits across four retained regimes, two optimizers and three rates per optimizer; rate selection used validation error. Twenty fresh paired seed blocks gave 320 fits. Every condition continued from 4,096 to 16,384 updates; minimum validation loss within each budget selected test endpoints. Selected and common rates coincided, so these are the same fits. A post hoc sensitivity restarted all conditions with log-conductance bounds $[-20, 20]$ and replayed the original $[-7, 7]$ bounds, yielding 640 additional trajectories on the same seed blocks. Intervals bootstrap whole seed blocks 20,000 times. At common parameter states, context-gradient cancellation is the norm of the probability-weighted sum of context-conditional distal gradient estimates divided by the weighted sum of their norms, before optimizer transformation. The original monotonic construction and every development outcome remain in Supplementary Figs. S50–S51.

Serial depth, sensor alignment and optimization

Physical-depth networks contain 64 somata and eight nonsomatic modules each. D1–D4 reallocate these modules across branching vectors $[8]$, $[2, 3]$, $[2, 1, 2]$ and $[1, 1, 2, 2]$. Contacts and resources are matched; serial composition changes. Grouped-point controls connect every module directly to the soma. Raw-additive controls retain the graph without division, while flexible MLP controls are approximately parameter matched.

Excitatory blocks carry class evidence multiplied by lognormal nuisance factors. Inhibitory blocks sense those factors with alignment α : zero removes trial-specific information while preserving the sensor distribution, and one reports the factor. The generator floors noisy input rates before multiplying them by gains. Nested, flat and local-ratio families distinguish distributed hierarchical support from generic factor cancellation and locally available ratios. Full generators, floors, sensor-placement controls and contact inventories are defined in Supplementary Section S5.

Ten paired seeds support each original cohort. The initial maximum is 180 epochs with patience thirty; validation selects model states. BP and LocalCA optimizer recipes differ in learning rates and parameter-group treatment, so their endpoint gaps are reported separately. Exact-path LocalCA computes the exact selected gradients. An audit of one hundred original trajectories verifies every configuration and final metric against its source: all fifty D3 fits across the five relevant credit/optimizer arms reached the epoch cap with validation loss still falling. The original depth contrasts are therefore fixed-budget outcomes. Sixty source-consistent restarts raise the maximum to 600 epochs, retaining all fifty D3 fits and ten exact-BP D1 references. Within each restart, minimum validation loss selects states in both budget windows. All D3 fits still reach the larger cap with decreasing validation loss. Complete histories, accuracy and loss contrasts, saved-state checks and source-concordance discrepancies are retained in Supplementary Section S5.

651 **Anatomical dictionaries and common-mode controls**

652 Observed MICrONS skeletons and mapped contacts define soma-rooted branch-to-branch segments. Initial eight
653 and disjoint 47-cell cohorts come from the same mouse; Pinky is a second mouse, with ten of twelve selected cells
654 meeting the inherited typed-contact criterion. Each subtree route contains the input-bearing descendants of its origin.
655 Contact labels are direct where available and otherwise proxy-based as specified in Supplementary Sections S6–S8.
656 Compression preserves path length but only approximates resistance; direct-label, mapping and geometry sensitivities
657 are retained.

658 The independent field consists of reciprocal passive-cable gradient changes induced by small focal shunts. Every
659 corrected dictionary receives one constant column and $K - 1$ spatial columns. Ancestry selection uses inhibitory
660 fraction times excitatory-area-weighted descendant fraction without field outcomes. Depth bins are reparameterized
661 with the same constant while preserving their span; random routes, independent site shuffles and degree–depth
662 surrogates provide controls. The weighted common component is removed before computing residual-energy capture.
663 A common-constrained SVD oracle supplies the rank-budget ceiling. Numerical rank, coverage and nonzero density
664 are reported separately; equal column count alone does not match those resources.

665 The original eight cells are post hoc development for common-mode correction. The protocol was frozen before
666 evaluating the new independent-generator outcomes in v661 and Pinky. Two hundred randomized controls are
667 averaged within each cell. At eight columns, all 47 v661 cells and eight Pinky cells qualify; other eligible budgets are
668 retained without substituting for missing eight-column outcomes. Paired cell-bootstrap intervals and Holm-adjusted
669 comparisons describe within-cohort consistency, not animal-population replication. The older route-generated
670 positive controls use separate fields and endpoint definitions (Supplementary Section S7).

671 **Focal shunting and the electrotonic boundary**

672 Up to sixteen focal sites per cell span depth quartiles and require at least three excitatory descendants and three
673 comparison sites. Normalized E/I scales are 0.35 with reversals one and -0.2 . Shunt doses are 0.25, 0.5, one and
674 two times local membrane-plus-synaptic conductance. A matched injection supplies first-order current $\kappa_k(E_I - V_k)$
675 without changing the conductance matrix. A separately solved somatic current restores baseline voltage and is held
676 fixed during differentiation. The squared-loss target is one normalized unit below baseline; finite differences check
677 the adjoint gradients.

678 Localization subtracts depth-matched off-route from descendant medians of $|\log[(|\gamma'_i| + \varepsilon)/(|\gamma_i| + \varepsilon)]|$, with
679 $\varepsilon = 10^{-15} \max(1, \max_i |\gamma_i|)$. Because this is unsigned, signed gradient changes are reported separately. Adjoint and
680 driving-force substitutions in $\gamma_i = q_i(E_E - V_i)$ distinguish transport from local voltage effects. Sites and relation-
681 reassignment templates are nested within cells. The passive ancestry-partition identity and its dictionary qualification
682 are derived in Supplementary Section S4.

683 Physical calibration uses cylindrical axial conductance $\pi r^2/(R_a \ell)$ and leak $2\pi r \ell/R_m$, expressed in nS, with
684 spherical soma area $4\pi r^2$. Here r and ℓ are segment radius and length, R_a axial resistivity and R_m specific membrane
685 resistance. Sweeps include $R_a = 100, 150, 250 \Omega \text{cm}$ and $R_m = 300\text{--}30,000 \Omega \text{cm}^2$, fixed and locally normalized
686 doses, and background conductance. The disjoint calibration cohort contains 235 eligible sites in 45 cells. Geometric
687 fallbacks and dose definitions are specified in Supplementary Section S9.

688 The weak-channel check uses $R_a = 150$, $R_m = 1,000$ in those units and background conductance equal to leak.
689 Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times
690 excitatory conductance. Sixty-four draws per cell passed the predefined steady-state checks. Responses use rank-one
691 updates of the fixed baseline Jacobian, including voltage-gate derivatives, followed by somatic state matching. They
692 are local linearizations rather than newly solved nonlinear equilibria at each dose (Supplementary Fig. S47).

693 **Measured-response prediction and geometric diagnostics**

694 The seven-target cohort includes thirteen eligible scans. Trial responses are the mean of the supplied ROI fluorescence
695 traces in `RoiResponseSeries` over each stimulus interval; our analysis did not deconvolve the traces or apply a
696 reliability correction. Repeated responses are averaged by stimulus condition for the ancestry-similarity analysis;

697 partial rank correlations control Euclidean separation and soma-to-contact path-length differences. Offline learning
698 holds out stimulus identities. Training-only percentile transforms produce nonnegative presynaptic proxies, and each
699 partner enters at its dominant mapped contact segment. A linear somatic readout predicts the standardized target
700 response. Only input-conductance credit differs across rules; readout and bias gradients remain exact.

701 Four-route dictionaries are selected without target responses and receive the projected baseline transfer profile, as
702 in Eq. 9. Realized supports are reconstructed directly from saved segment identifiers. The displayed matrix uses the
703 scan with median mapped-input count, with identifier-based tie breaking; all scan supports are retained. At common
704 exact-trained states, field and eligibility-weighted update reconstruction compare the fixed profile, per-trial field
705 projection and the best within-span update. These are oracle diagnostics rather than newly trained rules. The 130
706 exact re-executions retain their endpoint discrepancies.

707 Ridge regularization is selected within training data for every outer split, using the same measured predictors.
708 Observed-input removal, predictor noise and reliability masks are separate sensitivity analyses with no held-out
709 information used for selection. Ten split/initialization replicates are averaged within scans and then within seven
710 targets. Imposed-alignment quadratic fields and the six-animal external signed-credit reanalysis are separate datasets
711 and tests, fully described in Supplementary Sections S10–S12.

712 **Statistics and computational provenance**

713 Confirmatory choices precede outcome inspection but are not described as public preregistration. Pilots, post hoc
714 analyses and same-seed reruns are identified separately. Effect sizes accompany experiment-specific 95% intervals,
715 or adjusted 97.5% intervals where prespecified. Tests are two sided unless a directional protocol states otherwise.
716 Named multiplicity families and exact tie treatment are retained in Source Data; null results imply equivalence only
717 when a margin is tested.

718 Independent units are paired training seeds, reconstructed cells, postsynaptic targets or external animals. Sites,
719 contacts, scans, splits, checkpoints and randomized draws remain nested. Seed counts reflect the experiment’s
720 purpose; no prospective parametric power calculation was performed. Source hashes certify retained inputs and
721 implementations, not unavailable historical runtime states. The matched credit and new conductance studies recorded
722 Python 3.10.13, NumPy 2.2.6 and pandas 2.3.3. The new MNIST controls used PyTorch 2.9.1/CUDA 12.8 and
723 torchvision 0.24.1 with NumPy 2.2.6; the depth extension used Python 3.10 and PyTorch 2.9.1. The isolated reviewer
724 environment was validated with NumPy 1.26.4, SciPy 1.15.3, PyTorch 2.9.1, torchvision 0.24.1 and Matplotlib
725 3.10.9. Cohort-specific runtime records distinguish executed experiments from this installation check. Supplementary
726 Section S13 lists all exclusions, statistical families, numerical checks and release records.

727 OpenAI Codex assisted with code, statistical and consistency audits, figure preparation, literature checks and
728 language editing. The authors remain responsible for design, interpretation, numerical accuracy, citations and the
729 final manuscript.

730 **Ethics and data reuse**

731 No new experiments involving animals or human participants were performed. All biological analyses reused public
732 MICrONS, DANDI, and published animal source data. The source studies describe the original data collection and
733 institutional approvals [22, 38, 56].

734 **Data availability**

735 MICrONS anatomy and connectivity are available through the public CAVE datastack minnie65_public. The
736 primary cohort used materialization 1822, skeleton-service reconstructions, and the synapses_pni_2 table. The
737 disjoint sensitivity cohort used the public v661 static release. Secondary target-structure labels used only in the
738 full-contact analysis came from the in-development synapse_target_predictions_ssa_v2 table. These labels

are distributed only as derived data and remain subject to the source-data access terms. All principal structural findings were repeated without them.

The independent-animal replication used the MICrONS phase-1 layer-2/3 Pinky v185 fixed meshes, soma-valence table, and dense synapse table archived at Zenodo DOI 10.5281/zenodo.3710459 [57]. File sizes, Zenodo MD5 checksums, and local SHA-256 checksums are recorded in the supplied manifest. Volumetric imagery is archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as DANDI:000402. The supplied manifest records the exact asset and table identifiers, and the source publication describes the full data release [38].

Public benchmark datasets are available from their original repositories. The inhibitory-census tables and SWCs are available with the source record for Schneider-Mizell et al. [56]. The Francioni et al. workbook is available through the Source Data link associated with the Nature article [22]; its checksum and the sheet names used here are recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and non-restricted summaries are supplied with this manuscript. Authentication tokens and externally hosted raw caches are not redistributed.

Code availability

The accompanying reviewer software archive contains the complete training implementation, exact-gradient and perturbation analyses, versioned configurations, tests, figure scripts, and byte-identified snapshots of the critical training components under the MIT License. A separate Source Data archive contains the numerical data underlying every display item. These archives are available to editors and reviewers with the submission. The same version will be released publicly under the MIT License in a versioned repository and permanent archival record upon publication.

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Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the theory of local credit assignment, its implementation, and its interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.